Gemini 2.5: Pushing the Frontier with Advanced Reasoning, Multimodality, Long Context, and Next Generation Agentic Capabilities.

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In this report, we introduce the Gemini 2.X model family: Gemini 2.5 Pro and Gemini 2.5 Flash, as well as our earlier Gemini 2.0 Flash and Flash-Lite models. Gemini 2.5 Pro is our most capable model yet, achieving SoTA performance on frontier coding and reasoning benchmarks. In addition to its incredible coding and reasoning skills, Gemini 2.5 Pro is a thinking model that excels at multimodal understanding and it is now able to process up to 3 hours of video content. Its unique combination of long context, multimodal and reasoning capabilities can be combined to unlock new agentic workflows. Gemini 2.5 Flash provides excellent reasoning abilities at a fraction of the compute and latency requirements and Gemini 2.0 Flash and Flash-Lite provide high performance at low latency and cost. Taken together, the Gemini 2.X model generation spans the full Pareto frontier of model capability vs cost, allowing users to explore the boundaries of what is possible with complex agentic problem solving.

1. Introduction

We present our latest family of natively multimodal models with advanced reasoning through thinking, long context and tool-use capabilities: Gemini 2.5 Pro and 2.5 Flash and our earlier Gemini 2.0 Flash and Gemini 2.0 Flash-Lite models. Together these form a new family of highly-capable models representing our next generation of AI models, designed to power a new era of agentic systems. Building upon the foundation of the Gemini 1.5 series (Gemini Team, 2024), this Gemini 2.X generation brings us closer to the vision of a universal AI assistant (Hassabis, 2025).

The Gemini 2.X series are all built to be natively multimodal, supporting long context inputs of >1 million tokens and have native tool use support. This allows them to comprehend vast datasets and handle complex problems from different information sources, including text, audio, images, video and even entire code repositories. These extensive capabilities can also be combined to build complex agentic systems, as happened in the case of Gemini Plays Pokémon (Zhang, 2025). Different models in the series have different strengths and capabilities: (1) Gemini 2.5 Pro is our most intelligent thinking model, exhibiting strong reasoning and code capabilities. It excels at producing interactive web applications, is capable of codebase-level understanding and also exhibits emergent multimodal coding abilities. (2) Gemini 2.5 Flash is our hybrid reasoning model with a controllable thinking budget, and is useful for most complex tasks while also controlling the tradeoff between quality, cost, and latency. (3) Gemini 2.0 Flash is our fast and cost-efficient non-thinking model for everyday tasks and (4) Gemini 2.0 Flash-Lite is our fastest and most cost-efficient model, built for at-scale usage. A full comparison of the models in the Gemini 2.X model family is provided in Table 1. Taken together, the Gemini 2.X family of models cover the whole Pareto frontier of model capability vs cost, shifting it forward across a large variety of core capabilities, applications and use-cases see Figure 1.

The Gemini 2.5 family of models maintain robust Safety metrics while improving dramatically on helpfulness and general tone compared to their 2.0 and 1.5 counterparts. In practice, this means that

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Architecture gains are still left on the table

	Gemini 1.5 Flash	*Gemini 1.5 Pro*	**Gemini 2.0 Flash-Lite**	**Gemini 2.0 Flash**	**Gemini 2.5 Flash**	**Gemini 2.5 Pro**
Input modalities	Text, Image, Video, Audio	Text, Image, Video, Audio	Text, Image, Video, Audio	Text, Image, Video, Audio	Text, Image, Video, Audio	Text, Image, Video, Audio
Input length	1M	1M	1M	1M	1M	1M
Output modalities	Text	Text	Text	Text, Image*	Text, Audio*	Text, Audio
Output length	8K	8K	8K	8K	64K	64K
Thinking	No	No	No	Yes	Yes (controllable)	Yes (controllable)
Supports tool use?	No	No	No	Yes	Yes	Yes
Knowledge cutoff	November 2023	November 2023	June 2024	June 2024	January 2025	January 2025

Table 1 | Comparison of Gemini 2.X model family with Gemini 1.5 Pro and Flash. Tool use refers to the ability of the model to recognize and execute function calls (e.g., to perform web search, complete a math problem, execute code). **currently limited to Experimental or Preview, see Section 2.7. Information accurate as of publication date.*

the 2.5 models are substantially better at providing safe responses without interfering with important use cases or lecturing end users.

Our report is structured as follows: we begin by briefly describing advances we have made in model architecture, training and serving since the release of the Gemini 1.5 model. We then showcase the performance of the Gemini 2.5 models, including qualitative demonstrations of its abilities. We conclude by discussing the safety evaluations and implications of this model series.

2. Model Architecture, Training and Dataset

2.1. Model Architecture

The Gemini 2.5 models are sparse mixture-of-experts (MoE) (Clark et al., 2022; Du et al., 2021; Fedus et al., 2021; Jiang et al., 2024; Lepikhin et al., 2020; Riquelme et al., 2021; Roller et al., 2021; Shazeer et al., 2017) transformers (Vaswani et al., 2017) with native multimodal support for text, vision, and audio inputs. Sparse MoE models activate a subset of model parameters per input token by learning to dynamically route tokens to a subset of parameters (experts); this allows them to decouple total model capacity from computation and serving cost per token. Developments to the model architecture contribute to the significantly improved performance of Gemini 2.5 compared to Gemini 1.5 Pro (see Section 3). Despite their overwhelming success, large transformers and sparse MoE models are known to suffer from training instabilities (Chowdhery et al., 2022; Dehghani et al., 2023; Fedus et al., 2021; Lepikhin et al., 2020; Liu et al., 2020; Molybog et al., 2023; Wortsman et al., 2023; Zhai et al., 2023; Zhang et al., 2022). The Gemini 2.5 model series makes considerable progress in enhancing large-scale training stability, signal propagation and optimization dynamics, resulting in a considerable boost in performance straight out of pre-training compared to previous Gemini models.

Gemini 2.5 models build on the success of Gemini 1.5 in processing long-context queries, and incorporate new modeling advances allowing Gemini 2.5 Pro to surpass the performance of Gemini 1.5 Pro in processing long context input sequences of up to 1M tokens (see Table 3). Both Gemini 2.5 Pro and Gemini 2.5 Flash can process pieces of long-form text (such as the entirety of “Moby Dick” or “Don Quixote”), whole codebases, and long form audio and video data (see Appendix 8.5). Together with advancements in long-context abilities, architectural changes to Gemini 2.5 vision processing

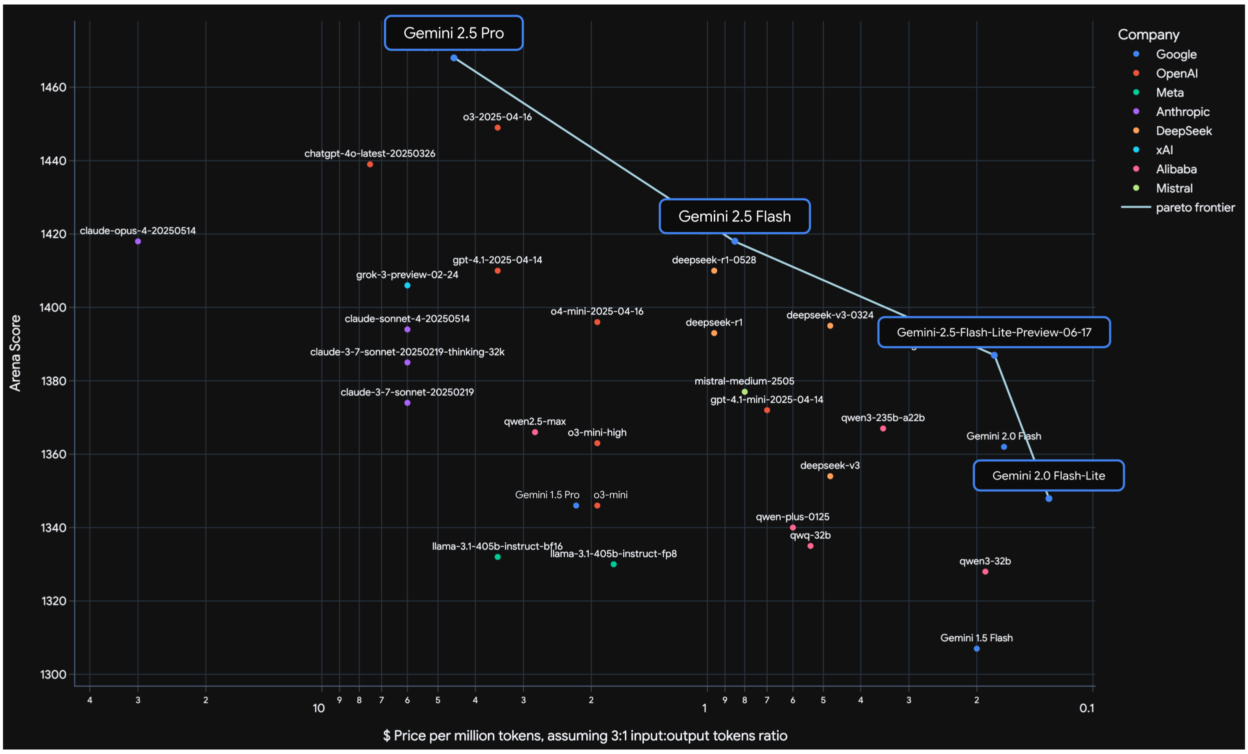

Figure 1 | Cost-performance plot. Gemini 2.5 Pro is a marked improvement over Gemini 1.5 Pro, and has an LMArena score that is over 120 points higher than Gemini 1.5 Pro. Cost is a weighted average of input and output tokens pricing per million tokens. Source: LMArena, imported on 2025-06-16.

lead to a considerable improvement in image and video understanding capabilities, including being able to process 3 hour long videos and the ability to convert demonstrative videos into interactive coding applications (see our recent blog post, (Baddepudi et al., 2025)).

The smaller models in the Gemini 2.5 series — Flash size and below — use distillation (Anil et al., 2018; Hinton et al., 2015), as was done in the Gemini 1.5 series (Gemini Team, 2024). To reduce the cost associated with storing the teacher's next token prediction distribution, we approximate it using a k-sparse distribution over the vocabulary. While this still increases training data throughput and storage demands by a factor of k, we find this to be a worthwhile trade-off given the significant quality improvement distillation has on our smaller models, leading to high-quality models with a reduced serving cost (see Figure 2).

2.2. Dataset

Our pre-training dataset is a large-scale, diverse collection of data encompassing a wide range of domains and modalities, which includes publicly available web-documents, code (various programming languages), images, audio (including speech and other audio types) and video, with a cutoff date as June 2024 for 2.0 and January 2025 for 2.5. Compared to the Gemini 1.5 pre-training dataset we also utilized new methods for improved data quality for both filtering, and deduplication. Our post-training dataset, like Gemini 1.5, consists of instruction tuning data that is carefully collected and vetted, is a collection of multimodal data with paired instructions and responses in addition to human preference and tool-use data.

Lol no real details provided here

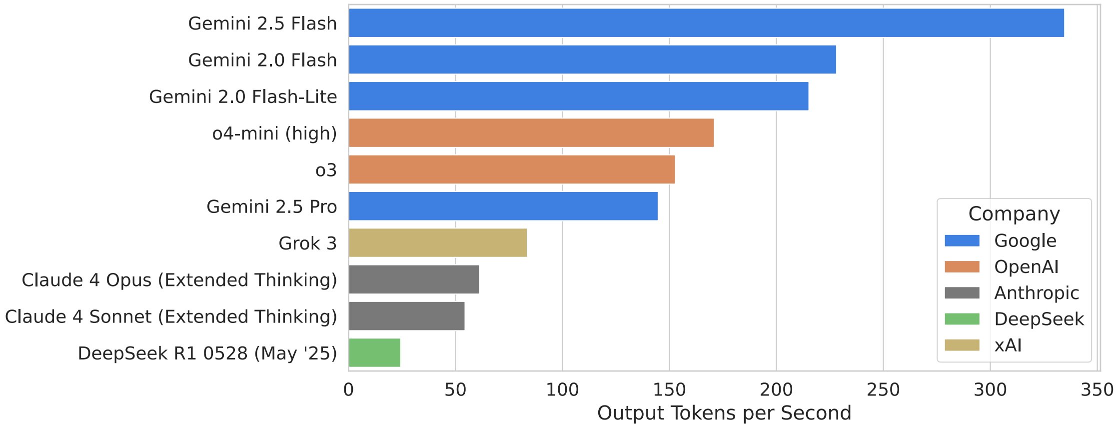

Figure 2 | Number of output tokens per second while generating (i.e. after the first chunk has been received from the API), for different models. Source: `ArtificialAnalysis.ai`, imported on 2025-06-15

2.3. Training Infrastructure

This model family is the first to be trained on TPUv5p architecture. We employed synchronous data-parallel training to parallelise over multiple 8960-chip pods of Google's TPUv5p accelerators, distributed across multiple datacenters.

Only DP Training ?

The main advances in software pre-training infrastructure compared with Gemini 1.5 were related to elasticity and mitigation of SDC (Silent Data Corruption) errors:

This is the privilege large labs have.

1. **Slice-Granularity Elasticity**: Our system now automatically continues training with fewer "slices" of TPU chips when there is a localized failure, and this reconfiguration results in tens of seconds of lost training time per interruption, compared with the 10 or more minute delay waiting for healthy machines to be rescheduled without elasticity; the system continues training at around 97% throughput while the failed slice is recovering. At the scale of this training run we see interruptions from hardware failures multiple times per hour, but our fault tolerance machinery is designed to tolerate the higher failure rates expected at much larger scales.
2. **Split-Phase SDC Detection**: On previous large-scale runs it could take many hours to detect and localize machines with SDC errors, requiring both downtime while debugging, and roll-back/replay of a large number of potentially corrupt training steps. We now use lightweight deterministic replay to immediately repeat any step with suspicious metrics, and compare per-device intermediate checksums to localize the root cause of any data corruption. Empirically, accelerators that start to exhibit intermittent SDCs are identified within a few minutes, and quickly excluded from the job. During this run, around 0.25% of steps were replayed due to suspected SDCs and 6% of these replays turned out to be genuine hardware corruption.

Both of the above techniques were relatively simple to implement due to the single controller design of the Pathways system (Barham et al., 2022), which allows all accelerators to be coordinated from a single python program with a global view of the system state. The controller can make use of parallel 'remote python' operations on TPU workers to monitor training metrics, track performance stragglers, and root-cause SDC errors.

Overall during the run, 93.4% of the time was spent performing TPU computations; the remainder was approximately spent half in elastic reconfigurations, and half in rare tail cases where

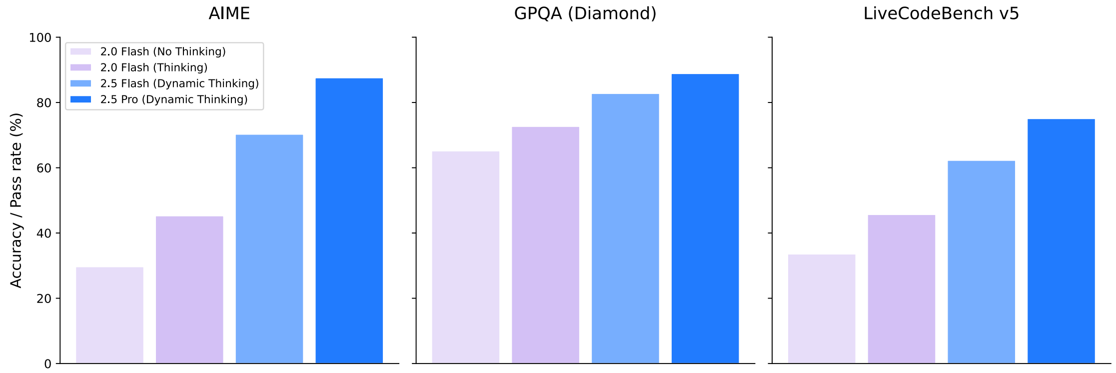

Figure 3 | Impact of "Thinking" on Gemini models performance.

elasticity failed. Around 4.5% of the computed steps were replays or rollbacks for model debugging interventions.

2.4. Post-training

Since the initial announcement of Gemini 1.5, significant advancements have been made in our post-training methodologies, driven by a consistent focus on data quality across the Supervised Fine-Tuning (SFT), Reward Modeling (RM), and Reinforcement Learning (RL) stages. A key focus has been leveraging the model itself to assist in these processes, enabling more efficient and nuanced quality control.

So vague haha

Furthermore, we have increased the training compute allocated to RL, allowing deeper exploration and refinement of model behaviors. This has been coupled with a focus on verifiable rewards and model-based generative rewards to provide more sophisticated and scalable feedback signals. Algorithmic changes to the RL process have also improved stability during longer training. These advancements have enabled Gemini 2.5 to learn from more diverse and complex RL environments, including those requiring multi-step actions and tool use. The combination of these improvements in data quality, increased compute, algorithmic enhancements, and expanded capabilities has contributed to across-the-board performance gains (as described in Section 3) , notably reflected in the significant increase in the model's LMArena ELO scores, with both Gemini 2.5 Flash and Pro gaining more than 120 points over their Gemini 1.5 counterparts (122 for Gemini 2.5 Pro and 111 for Gemini 2.5 Flash, see Figure 1), along with significant improvements on several other frontier benchmarks.

2.5. Thinking

Past Gemini models produce an answer immediately following a user query. This constrains the amount of inference-time compute (Thinking) that our models can spend reasoning over a problem. Gemini Thinking models are trained with Reinforcement Learning to use additional compute at inference time to arrive at more accurate answers. The resulting models are able to spend tens of thousands of forward passes during a "thinking" stage, before responding to a question or query.

Our training recipe has evolved from the original experimental thinking model, Gemini 2.0 Flash Thinking (launched in December 2024), which excelled in mathematics and coding, to the Gemini 2.5 Thinking series, which incorporates Thinking natively across all domains. The result is a single model that can achieve stronger reasoning performance across the board, and is able to scale up its performance further as a function of inference time (see Figure 3 for an example of the impact of Thinking).

Model	AI Studio model ID
Gemini 1.5 Flash	gemini-1.5-flash
Gemini 1.5 Pro	gemini-1.5-pro
Gemini 2.0 Flash-Lite	gemini-2.0-flash-lite
Gemini 2.0 Flash	gemini-2.0-flash
Gemini 2.5 Flash	gemini-2.5-flash
Gemini 2.5 Pro	gemini-2.5-pro

Table 2 | Mapping of Gemini model names to AI Studio API model IDs.

We integrated Thinking with other Gemini capabilities, including native multimodal inputs (images, text, video, audio) and long context (1M+ tokens). For any of these capabilities, the model decides for itself how long to think before providing an answer. We also provide the ability to set a Thinking budget, constraining the model to respond within a desired number of tokens. This allows users to trade off performance with cost.

The Gemini 2.5 Thinking models are our most well-rounded reasoning models to date.

2.6. Capability-specific improvements

While most of the changes made to our training architecture and recipe since Gemini 1.5 have resulted in improvements across all capabilities, we have also made changes that have resulted in some capability-specific wins. We will now discuss these for code, factuality, long context, multilinguality, audio, video, and agentic use cases (with a particular focus on Gemini Deep Research).

Code Pre-training gains still left

Gemini 2.0 and 2.5 represent a strategic shift of our development priorities towards delivering tangible real-world value, empowering users to address practical challenges and achieve development objectives within today's complex, multimodal software environments. To realize this, concerted efforts have been undertaken across both pre-training and post-training phases since Gemini 1.5. In pre-training, we intensified our focus on incorporating a greater volume and diversity of code data from both repository and web sources into the training mixture. This has rapidly expanded coverage and enabled the development of more compute-efficient models. Furthermore, we have substantially enhanced our suite of evaluation metrics for assessing code capabilities aligned with downstream use cases, alongside improving our ability to accurately predict model performance. During post-training, we developed novel training techniques incorporating reasoning capabilities and curated a diverse set of engineering tasks, with the aim to equip Gemini with effective problem-solving skills crucial for addressing modern engineering challenges. Key applications demonstrating these advancements include IDE functionalities, code agent use cases for complex, multi-step operations within full repositories, and multimodal, interactive scenarios such as end-to-end web and mobile application development. Collectively, these efforts have yielded broad and significant improvements in Gemini's coding capabilities. This progress is evidenced by superior performance on established benchmarks: performance on LiveCodeBench increased from 30.5% for Gemini 1.5 Pro to 69.0% for Gemini 2.5 Pro, while that for Aider Polyglot went from 16.9% to 82.2%. Performance on SWEBench-verified went from 34.2% to 67.2%, see Table 3 and Figure 4 in Section 3.2. Furthermore, Gemini 2.5 Pro obtained an increase of over 500 Elo over Gemini 1.5 Pro on the LMArena WebDev Arena (Chiang et al., 2024; LMArena Team, 2025), resulting in meaningful enhancements in practical applications, including UI and web application development (Doshi, 2025a), and the creation of sophisticated agentic workflows (Kilpatrick, 2025).

lol

Factuality

Within the context of generative models, ensuring the factuality of model responses to information-seeking prompts remains a core pillar of Gemini model development. With Gemini 1.5, our research was concentrated on enhancing the model's world knowledge and its ability to provide answers faithfully grounded in the context provided within the prompt. This effort culminated in the December 2024 release of FACTS Grounding (Jacovi et al., 2025), now an industry-standard benchmark for evaluating an LLM's capacity to generate responses grounded in user-provided documents. With Gemini 2.0 and 2.5, we have significantly expanded our scope to address multimodal inputs, long-context reasoning, and model-retrieved information. At the same time, the landscape and user expectations for factuality have evolved dramatically, shaped in part by Google's deployment of AI Overviews and AI Mode (Stein, 2025). To meet these demands, Gemini 2.0 marked a significant leap as our first model family trained to natively call tools like Google Search, enabling it to formulate precise queries and synthesize fresh information with sources. Building on this, Gemini 2.5 integrates advanced reasoning, allowing it to interleave these search capabilities with internal thought processes to answer complex, multi-hop queries and execute long-horizon tasks. The model has learned to use search and other tools, reason about the outputs, and issue additional, detailed follow-up queries to expand the information available to it and to verify the factual accuracy of the response. Our latest models now power the experiences of over 1.5B monthly active users in Google's AI Overviews and 400M users in the Gemini App. These models exhibit state-of-the-art performance across a suite of factuality benchmarks, including SimpleQA for parametric knowledge (Wei et al., 2024), FACTS Grounding for faithfulness to provided documents (Jacovi et al., 2024, 2025), and the Vectara Hallucination Leaderboard (Hughes et al., 2023), cementing Gemini as the model of choice for information-seeking demands.

Long context

Modeling and data advances helped us improve the quality of our million-length context, and we reworked our internal evaluations to be more challenging to help steer our modeling research. When hill-climbing, we targeted challenging retrieval tasks (like LOFT (Lee et al., 2024)), long-context reasoning tasks (like MRCR-V2 (Vodrahalli et al., 2024)), and multimodal tasks (like VideoMME (Fu et al., 2025)). According to the results in Table 6, the new 2.5 models improve greatly over previous Gemini 1.5 models and achieve state-of-the-art quality on all of those. An example showcasing these improved capabilities for video recall can be seen in Appendix 8.5, where Gemini 2.5 Pro is able to consistently recall a 1 sec visual event out of a full 46 minutes video.[1]

Multilinguality

Gemini's multilingual capabilities have also undergone a profound evolution since 1.5, which already encompassed over 400 languages via pretraining. This transformation stems from a holistic strategy, meticulously refining pre- and post-training data quality, advancing tokenization techniques, innovating core modeling, and executing targeted capability hillclimbing. The impact is particularly striking in Indic and Chinese, Japanese and Korean languages, where dedicated optimizations in data quality and evaluation have unlocked dramatic gains in both quality and decoding speed. Consequently, users benefit from significantly enhanced language adherence, responses designed to faithfully respect the requested output language, and a robust improvement in generative quality and factuality across languages, solidifying Gemini's reliability across diverse linguistic contexts.

[1] For further discussion on long context capabilities, challenges, and future outlook, the Release Notes podcast episode "Deep Dive into Long Context" provides additional insights and discussion: `https://youtu.be/NHMJ9mqKeMQ`.

Only 66 tokens for video. Slowing reaching C4P level of tokens.

Audio

While Gemini 1.5 was focused on native audio understanding tasks such as transcription, translation, summarization and question-answering, in addition to understanding, Gemini 2.5 was trained to perform audio generation tasks such as text-to-speech or native audio-visual to audio out dialog. To enable low-latency streaming dialog, we incorporated causal audio representations that also allow streaming audio into and out of Gemini 2.5. These capabilities derive from an increased amount of pre-training data spanning over 200 languages, and development of improved post-training recipes. Finally, through our improved post-training recipes, we have integrated advanced capabilities such as Thinking, affective dialog, contextual awareness and tool use into Gemini's native audio models.

Video

We have significantly expanded both our pretraining and post-training video understanding data, improving the audio-visual and temporal understanding capabilities of the model. We have also trained our models so that they perform competitively with 66 instead of 258 visual tokens per frame, enabling using about 3 hours of video instead of 1h within a 1M tokens context window[2]. Two new applications that were not previously possible, but that have been unlocked as a result of these changes are: creating an interactive app (such as a quiz to test students' understanding of the video content) from a video and creating a p5.js animation to show the key concepts from the video. Our recent blog post (Baddepudi et al., 2025) shows examples of these applications.

Gemini as an Agent: Deep Research

Gemini Deep Research is an agent built on top of the Gemini 2.5 Pro model designed to strategically browse the web and provide informed answers to even the most niche user queries. The agent is optimized to perform task prioritization, and is also able to identify when it reaches a dead-end when Browse. We have massively improved the capabilities of Gemini Deep Research since its initial launch in December 2024. As evidence of that, performance of Gemini Deep Research on the Humanity's Last Exam benchmark (Phan et al., 2025) has gone from 7.95% in December 2024 to the **SoTA score of 26.9% and 32.4% with higher compute** (June 2025).

HLE is going down!

2.7. The path to Gemini 2.5

On the way to Gemini 2.5 Pro, we experimented with our training recipe, and tested a small number of these experimental models with users. We have already discussed Gemini 2.0 Flash Thinking (see Section 2.5). We will now discuss some of the other models briefly.

Gemini 2.0 Pro

In February 2025, we released an experimental version of Gemini 2.0 Pro. At the time, it had the strongest coding performance of any model in the Gemini model family, as well as the best understanding and world knowledge. It also came with our largest context window at 2 million tokens, which enabled it to comprehensively analyze and understand vast amounts of information. For further information about Gemini 2.0 Pro, please see our earlier blog posts (Kavukcuoglu, 2025; Mallick and Kilpatrick, 2025).

[2]this is referred to as low media resolution in the API: `https://ai.google.dev/api/generate-content#MediaResolution`

Gemini 2.0 Flash Native Image Generation Model

In March 2025, we released an experimental version of Gemini 2.0 Flash Native Image Generation. It has brought to the users new capabilities as a result of a strong integration between the Gemini model and image-generation capabilities, enabling new experiences related to image generation & image editing via natural-language prompting. Capabilities such as multi-step conversational editing or interleaved text-image generation are very natural in such a setting, and horizontal transfer related to multi-language coverage immediately allowed such experiences to happen across all the languages supported by the Gemini models. Native image generation turns Gemini into a multimodal creation partner and enables Gemini to express ideas through both text and images, and seamlessly move between the two. For further information about Gemini 2.0 Flash Native Image Generation, please see our earlier blog posts (Kampf and Brichtova, 2025; Sharon, 2025)

Gemini 2.5 Audio Generation

Nice 2.5 Pro used in NotebookLM

With Gemini 2.5, the Controllable TTS and Native Audio Dialog capabilities are available as separate options on AI Studio (Generate Media and Stream sections respectively). Our Gemini 2.5 Preview TTS Pro and Flash models support more than 80 languages with the speech style controlled by a free formatted prompt which can specify style, emotion, pace, etc, while also being capable of following finer-grained steering instructions specified in the transcript. Notably, Gemini 2.5 Preview TTS can generate speech with multiple speakers, which enables the creation of podcasts as used in NotebookLM Audio Overviews (Wang, 2024). Our Gemini 2.5 Flash Preview Native Audio Dialog model uses native audio generation, which enables the same level of style, pacing and accent control as available in our controllable TTS offering. Our dialog model supports tool use and function calling, and is available in more than 24 languages. With native audio understanding and generation capabilities, it can understand and respond appropriately to the user's tone. This model is also capable of understanding when to respond to the user, and when not to respond, ignoring background and non-device directed audio. Finally, we also offer an advanced 'Thinking' variant that effectively handles more complex queries and provides more robust and reasoned responses in exchange for some additional latency.

Some VAD type filter?

Thinking in audio !!

Gemini 2.5 Flash-Lite

In June 2025, we released an experimental version of Gemini 2.5 Flash-Lite (`gemini-2.5-flash-lite-preview-06-17`). It comes with the same capabilities that make Gemini 2.5 helpful, including the ability to turn thinking on at different budgets, connecting to tools like Google Search and code execution, multimodal input and a 1 million-token context length. Our goal was to provide an economical model class which provides ultra-low-latency capabilities and high throughput per dollar, echoing the initial release of 2.0 Flash-Lite (Google DeepMind, 2025b; Mallick and Kilpatrick, 2025).

Gemini 2.5 Pro Deep Think

To advance Gemini's capabilities towards solving hard reasoning problems, we developed a novel reasoning approach, called Deep Think, that naturally blends in parallel thinking techniques during response generation. Deep Think enables Gemini to creatively produce multiple hypotheses and carefully critique them before arriving at the final answer, achieving state-of-the-art performances in challenging benchmarks such as Olympiad math (USAMO 2025), competitive coding (LiveCodeBench), and multimodality (MMMU), see more details at (Doshi, 2025b). We announced Gemini 2.5 Deep Think at Google I/O and launched an experimental version to trusted testers and advanced users in June 2025.

⇒ Parallel thinking approachs as covered in other papers

3. Quantitative evaluation

We will now examine the performance of the Gemini 2.X model family across a wide range of academic benchmarks. We will first compare the performance of the Gemini 2.X models to the earlier Gemini 1.5 Pro and Flash models, before we compare the performance of Gemini 2.5 Pro to other available large language models.

With web-scale pre-training of AI models, coupled with the post-training techniques that allow policy and reward models to leverage public benchmarks, avoiding leaks and biases in the data used for pre- and post-training is a persistent challenge. In the development of the Gemini 2.5 series, in addition to the standard n-gram based decontamination we used in Gemini 1.5, we also employed semantic-similarity and model based decontamination procedures to help mitigate evaluation set leakage. To move beyond the reliance on training set decontamination, we also continue reporting on internally developed non-public benchmarks, such as HiddenMath.

3.1. Methodology

In Table 3, we compare the performance of Gemini 2.5 models to the Gemini 1.5 models, while in Table 4, we compare the performance of Gemini 2.5 Pro to that of other large language models.

Gemini results: All Gemini scores are pass@1, and are "single attempt" settings unless otherwise specified. In the "single attempt" setting, no majority voting or parallel test-time compute is permitted. "multiple attempts" settings allow test-time selection of the candidate answer. All Gemini evaluations are run with the AI Studio API for the model id that we provide in Table 2, with default sampling settings. To reduce variance, we average over multiple trials for smaller benchmarks. Aider Polyglot score is the pass rate average of 3 trials. Vibe-Eval results are reported using Gemini as a judge.

Non-Gemini results: All the results for non-Gemini models are sourced from providers' self reported numbers unless mentioned otherwise. All "SWE-bench Verified" numbers follow official

Hopefully they set the same 'default' settings when evaluating other models

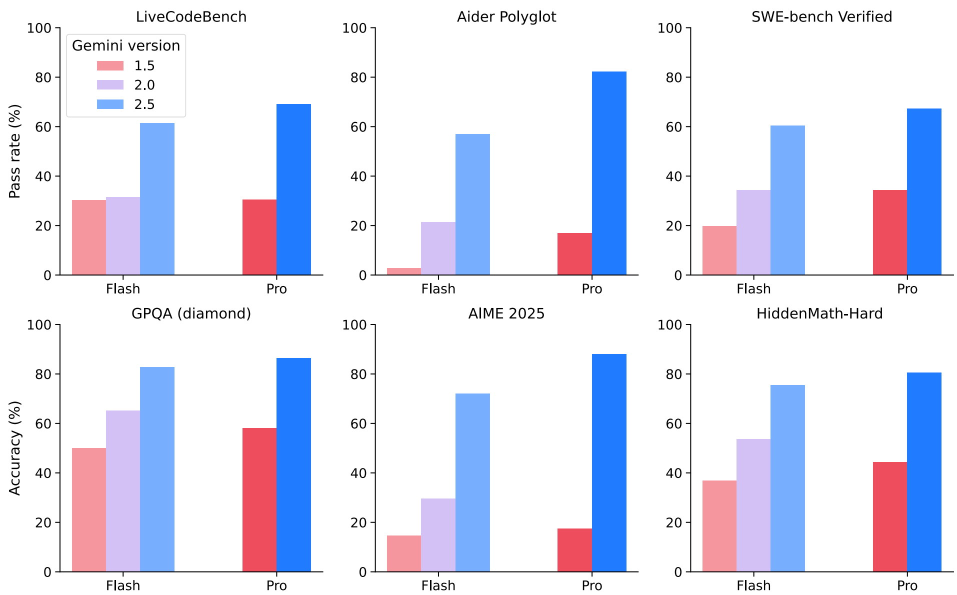

Figure 4 | Performance of Gemini 2.X models at coding, math and reasoning tasks in comparison to previous Gemini models. SWE-bench verified numbers correspond to the "multiple attempts" setting reported in Table 3.

provider reports, which means that they are computed using different scaffoldings and infrastructure, and aren't directly comparable. Single attempt vs multiple attempts: When two numbers are reported for the same eval higher number uses majority voting with n=64 for Grok models and internal scoring with parallel test time compute for Anthropic models.

Where provider numbers are not available we report numbers from leaderboards reporting results on these benchmarks. Results for Humanity's Last Exam results are sourced from the main leaderboard and the text-only leaderboard for DeepSeek (indicated with a ⋄ in Table 4) and in the case of the Gemini 2.0 models, these results are on an earlier HLE dataset (indicated with a † in Table 3). Results on LiveCodeBench results are taken from (1/1/2025 - 5/1/2025) in the UI. Aider Polyglot numbers come from the main leaderboard and results for SimpleQA come from this repo where available. Results on FACTS Grounding come from Kaggle. In the case of LOFT and MRCR-V2, we report results on both the 128k context length variant, as well as the 1M context length variant. In the 128k context length variant, we measure performance on contexts up to 128k, while for the 1M context length variant, we report performance on context lengths of exactly 1M.

More details on all benchmarks, including subsets and how/where scores were obtained can be found in Table 11 in Appendix 8.1.

3.2. Core capability quantitative results

As can be seen in Table 3, and Figure 4, the Gemini 2.5 models excel at coding tasks such as LiveCodeBench, Aider Polyglot and SWE-bench Verified, and represent a marked improvement over previous models.

In addition to coding performance, Gemini 2.5 models are noticeably better at math and reasoning tasks than Gemini 1.5 models: performance on AIME 2025 is 88.0% for Gemini 2.5 Pro compared to 17.5% for Gemini 1.5 Pro, while performance on GPQA (diamond) went from 58.1% for Gemini 1.5 Pro to 86.4%. Similarly, Image understanding has increased significantly.

It is also interesting to note that the Gemini 2.5 Flash model has become the second most capable model in the Gemini family, and has overtaken not just previous Flash models, but also the Gemini 1.5 Pro model released one year ago.

Capability	Benchmark		Gemini 1.5 Flash	Gemini 1.5 Pro	Gemini 2.0 Flash-Lite	Gemini 2.0 Flash	Gemini 2.5 Flash	Gemini 2.5 Pro
Code	LiveCodeBench		30.3%	29.7%	29.1%	29.1%	55.4%	**69.0%**
	Aider Polyglot		2.8%	16.9%	10.5%	21.3%	56.7%	**82.2%**
	SWE-bench Verified	*single attempt*	9.6%	22.3%	12.5%	21.4%	48.9%	**59.6%**
		multiple attempts	19.7%	34.2%	23.1%	34.2%	60.3%	**67.2%**
Reasoning	GPQA (diamond)		50.0%	58.1%	50.5%	65.2%	82.8%	**86.4%**
	Humanity's Last Exam	*no tools*	-	4.6%	4.6% †	5.1% †	11.0%	**21.6%**
Factuality	SimpleQA		8.6%	24.9%	16.5%	29.9%	26.9%	**54.0%**
	FACTS Grounding		82.9%	80.0%	82.4%	84.6%	85.3%	**87.8%**
Multilinguality	Global MMLU (Lite)		72.5%	80.8%	78.0%	83.4%	88.4%	**89.2%**
	ECLeKTic		16.4%	27.0%	27.7%	33.6%	36.8%	**46.8%**
Math	AIME 2025		14.7%	17.5%	23.8%	29.7%	72.0%	**88.0%**
	HiddenMath-Hard		36.8%	44.3%	47.4%	53.7%	75.5%	**80.5%**
Long-context	LOFT (hard retrieval)	*≤128K*	67.3%	75.9%	50.7%	58.0%	82.1%	**87.0%**
		1M	36.7%	47.1%	7.6%	7.6%	58.9%	**69.8%**
	MRCR-V2 (8-needle)	*≤128K*	18.4%	26.2%	11.6%	19.0%	54.3%	**58.0%**
		1M	10.2%	12.1%	4.0%	5.3%	**21.0%**	16.4%
Image Understanding	MMMU		58.3%	67.7%	65.1%	69.3%	79.7%	**82.0%**
	Vibe-Eval (Reka)		52.3%	55.9%	51.5%	55.4%	65.4%	**67.2%**
	ZeroBench		0.5%	1.0%	0.75%	1.25%	2.0%	**4.5%**
	BetterChartQA		59.0%	65.8%	52.3%	57.8%	67.3%	**72.4%**

Table 3 | Evaluation of Gemini 2.5 family across a wide range of core capability benchmarks and in comparison to Gemini 1.5 models. Please see Tables 5 and 6 for audio and video evaluations. See Table 11 Appendix 8.1 for benchmarks and evaluation details.

Still the only lab supporting 1M tokens.

3.3. Evaluation of Gemini 2.5 Pro against other large language models

Relative to other large language models that are available (see Table 4), Gemini achieves the SoTA score on the Aider Polyglot coding task. Gemini also achieves the highest score on Humanity's Last Exam, GPQA (diamond), and on the SimpleQA and FACTS Grounding factuality benchmarks out of all of the models examined here. Gemini also continues to stand out for achieving the SoTA score on both the LOFT and MRCR long-context tasks at 128k context, and is the only one, amongst the models examined in the above table, to support context lengths of 1M+ tokens.

Not all of the models shown in Table 4 have native support for multimodal inputs. As such, we compare against a different set of models for audio and video understanding.

Audio Understanding

In Table 5, we showcase the performance of the Gemini 2.5 model family at audio understanding, and compare the performance of these models to earlier Gemini models, as well as to GPT models. Gemini 2.5 Pro demonstrates state-of-the-art audio understanding performance as measured by public benchmarks for ASR and AST, and compares favorably to alternatives under comparable testing conditions (using the same prompts and inputs).

Video Understanding

In Table 6, we show the performance of Gemini 2.5 models at video understanding. As can be seen, Gemini 2.5 Pro achieves state-of-the-art performance on key video understanding benchmarks, surpassing recent models like GPT 4.1 under comparable testing conditions (same prompt and video

Capability	Benchmark		Gemini 2.5 Pro	o3 *high*	o4-mini *high*	Claude 4 Sonnet	Claude 4 Opus	Grok 3 Beta *Extended Thinking*	DeepSeek R1 0528
Code	LiveCodeBench		69.0%	72.0%	**75.8%**	48.9%	51.1%	–	70.5%
	Aider Polyglot		**82.2%**	79.6%	72.0%	61.3%	72.0%	53.3%	71.6%
	SWE-bench Verified	*single attempt*	59.6%	69.1%	68.1%	**72.7%**	72.5%	-	-
		multiple attempts	67.2%	-	-	**80.2%**	79.4%	-	57.6%
Reasoning	GPQA (diamond)	*single attempt*	**86.4%**	83.3%	81.4%	75.4%	79.6%	80.2%	81.0%
	Humanity's Last Exam	*no tools*	**21.6%**	20.3%	18.1%	7.8%	10.7%	-	14.0% ⋄
Factuality	SimpleQA		**54.0%**	48.6%	19.3%	-	-	43.6%	27.8%
	FACTS Grounding		**87.8%**	69.9%	62.1%	79.1%	77.7%	74.8%	82.4%
Math	AIME 2025		88.0%	88.9%	**92.7%**	70.5%	75.5%	77.3%	87.5%
Long-context	LOFT (hard retrieval)	*≤128K*	**87.0%**	77.0%	60.5%	81.6%	-	73.1%	-
		1M	**69.8%**	-	-	-	-	-	-
	MRCR-V2 (8-needle)	*≤128K*	**58.0%**	57.1%	36.3%	39.1%	*16.1%**	34.0%	-
		1M	**16.4%**	-	-	-	-	-	-
Image Understanding	MMMU	*single attempt*	82.0%	**82.9%**	81.6%	74.4%	76.5%	76.0%	*No MM support*

Table 4 | Performance comparison of Gemini 2.5 Pro with other large language models on different capabilities. Please see Tables 5 and 6 for audio and video evaluations. See Table 11 for benchmarks and evaluation details. **: with no thinking and API refusals*

Benchmark	**Gemini 1.5 Flash**	**Gemini 1.5 Pro**	**Gemini 2.0 Flash-Lite**	**Gemini 2.0 Flash**	**Gemini 2.5 Flash**	**Gemini 2.5 Pro**	**GPT-4o mini Audio Preview**	**GPT 4o Audio Preview**	**GPT 4o transcribe**
FLEURS *(53 lang, WER ↓)*	12.71	7.14	9.60	9.04	9.95	**6.66**	19.52	12.16	8.17
CoVoST2 *(21 lang, BLEU ↑)*	34.81	37.53	34.74	36.35	36.15	**38.48**	29.5	35.89	–

Table 5 | Performance comparison of Gemini 2.5 models to earlier Gemini models, as well as to GPT models for audio understanding. Note that for GPT models, metrics may differ from those previously reported due to differing eval methodologies. See Table 11 for benchmarks and evaluation details.

Modalities	**Benchmark**	**Gemini 1.5 Flash**	**Gemini 1.5 Pro**	**Gemini 2.0 Flash-Lite**	**Gemini 2.0 Flash**	**Gemini 2.5 Flash**	**Gemini 2.5 Pro**	**OpenAI GPT 4.1**
visual-only	ActivityNet-QA	56.2	57.3	55.3	56.4	65.1	**66.7**	60.4
	EgoTempo	34.5	36.3	30.1	39.3	36.7	**44.3**	40.3
	Perception Test	66.5	69.4	67.5	68.8	75.1	**78.4**	64.8
	QVHighlights	64.4	68.7	25.7	63.9	52.4	**75.0**	71.4
	VideoMMMU	64.8	70.4	64.3	68.5	79.2	**83.6**	60.9
	1H-VideoQA	61.9	72.2	55.6	67.5	67.5	**81.0**	56.8
audio + visual	LVBench	61.9	65.7	52	61.8	62.7	**78.7**	63.4
	VideoMME	70.4	73.2	62.1	72.8	75.5	**84.3**	72.0
	VATEX	56.9	55.5	58.5	56.9	65.2	**71.3**	64.1
	VATEX-ZH	46.2	52.2	43.2	48.5	43.9	**59.7**	48.7
	YouCook2 Cap	153.2	170.0	78.6	129.0	177.6	**188.3**	127.6
visual + subtitles	Minerva	49.6	52.8	46.8	52.4	60.7	**67.6**	54.0
	Neptune	78.7	82.7	81.5	83.1	84.3	**87.3**	85.2
audio+visual+ subtitles	VideoMME	77.3	79.8	72.5	78.8	81.5	**86.9**	79.6

Table 6 | Evaluation of Gemini 2.5 vs. prior models and GPT 4.1 on video understanding benchmarks. Performance is measured by string-match accuracy for multiple-choice VideoQA, LLM-based accuracy for open-ended VideoQA, R1@0.5 for moment retrieval and CIDEr for captioning. See Table 11 for benchmarks and evaluation details.

frames). For cost-sensitive applications, Gemini 2.5 Flash provides a highly competitive alternative.

4. Example use cases

4.1. Gemini Plays Pokemon

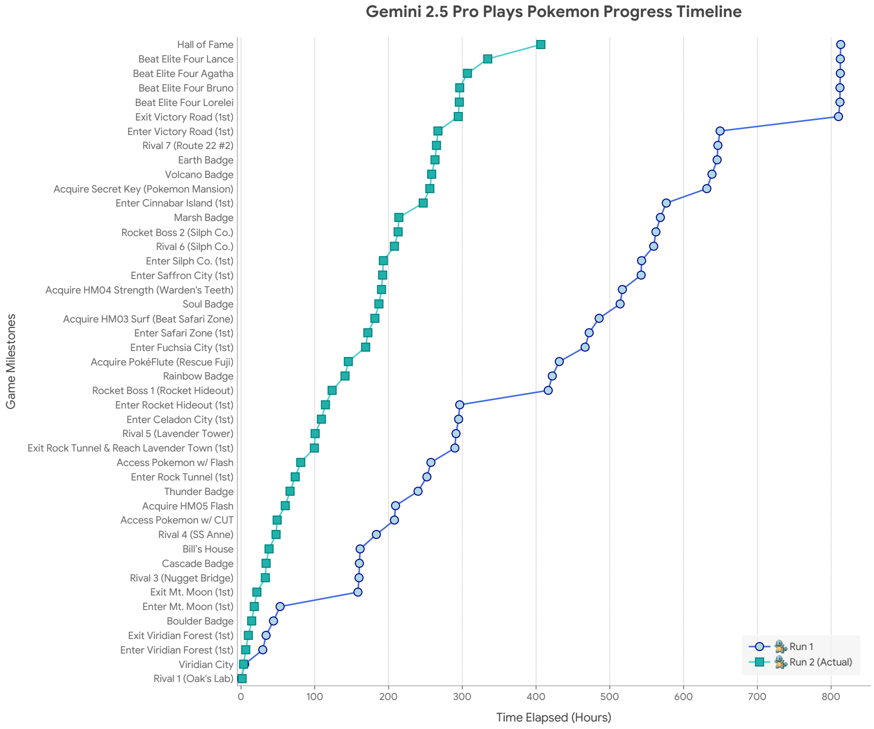

Figure 5 | Progression of the Gemini Plays Pokémon agent through the game, across two runs. Run 1 was the development run where changes to the harness were performed. Run 2 is the fully autonomous run with the final fixed scaffold. Both runs have the same starter (Squirtle). The events are ordered on the y-axis by the order they happened, following the order of Run 2 when there is a conflict. Notably, the GPP agent additionally went through the difficult (and optional) Seafoam Islands dungeon in Run 2, while in Run 1, GPP reached Cinnabar Island via Pallet Town and Route 21.

On March 28, 2025, an independent developer not affiliated with Google, Joel Zhang, set up a Twitch stream (Gemini Plays Pokémon, or GPP) for Gemini 2.5 Pro (Gemini 2.5 Pro Exp 03-25) to play Pokémon Blue on stream (Zhang, 2025) as an experiment to better understand how well the model was capable of playing Pokémon (in a similar spirit to Claude Plays Pokémon, see Anthropic (2025)). In this initial run through the game, the goal was to live-stream the development process of an agentic harness capable of playing the full game (and in particular the minimal transformation of vision to text necessary to do so), see Figure 13 for a description of the final agent setup. As such, over the course of the run, modifications were made to the setup as difficulties arose, providing a deeply interesting lens via which to analyze some of the qualitative improvements that the 2.5 Pro model has made, particularly in the regimes of solving long reasoning problems and agentic capabilities over extended time horizons. Around 1 month later, on May 2, 2025, Gemini 2.5 Pro completed the game after 813 hours and entered the Hall of Fame to become the Pokémon League Champion! On May 22, 2025, GPP began a fully autonomous 2nd run through the game with Gemini 2.5 Pro (*Gemini 2.5 Pro Preview 05-06*) with the finalized fixed agentic harness, and progressed through the game considerably faster, completing the game in 406.5 hours (nearly exactly half the time of the first run).

See Figure 5 for a timeline of GPP's progress through major game milestones to game completion. We report # hours to each milestone in order to normalize for the amount of time models take per action. See Appendix 8.2 for more figures.

Capabilities assessment

Gemini 2.5 Pro showcased many impressive capabilities associated with reasoning and long-term planning while playing Pokémon. For more examples, see Appendix 8.2.

Long Context Agentic Tooling Within the agent scaffolding, GPP has access to two agentic tools (see Figure 13). These prompted versions of Gemini 2.5 Pro, hereafter `pathfinder` and `boulder_puzzle_strategist`, have been able to:

1. Solve complex spinner puzzles in one shot (for instance in Rocket Hideout),
2. Solve the step-constrained multi-map puzzle of the Safari Zone,
3. Find long pathways through complex mazes like Route 13,
4. Solve boulder puzzles across long distances in Victory Road and the Seafoam Islands.

Each task requires reasoning over a long context - the `pathfinder` model would often have to reason over contexts of 100K+ tokens, and find paths up to 50 actions in length (in the extreme case, paths consisting of up to 150 actions have also been found!).

Long Horizon Task Coherence While Gemini 2.5 Pro is impressive in a more local sense, the agent also exhibited remarkable long-term task coherence in achieving global, high-level goals in the face of real and hallucinated setbacks towards making forward progress. Because the agent is able to change goals at will, and will generally follow those goals as long as needed, it is extremely impressive that the agent can satisfy numerous requirements for tactical, necessary goals, such as acquiring Hidden Moves, as well as maintain enough strategic task coherence to beat the entire game and become the Pokémon Champion.

Where does 2.5 Pro struggle while playing Pokémon?

In addition to more standard hallucination issues (which interestingly were plausibly reduced in Run 2 by explicitly prompting the model to act as a player completely new to the game, see Appendix 8.2 for more details), there are a few particular points of struggle we would like to emphasize.

Screen reading While obtaining excellent benchmark numbers on real-world vision tasks, 2.5 Pro struggled to utilize the raw pixels of the Game Boy screen directly, though it could occasionally take cues from information on the pixels. As a result, it was necessary for the required information from the screen to be translated into a text format in the agent framework, using information from the game's RAM state. During one portion of the game, the developer tested an ablation where all vision was completely removed from the model context – the model was able to function roughly as well as without the vision information, suggesting that most of the performance does not significantly depend on the visual input.

Visual input not that important

Long Context Reasoning Gemini 2.5 Pro's state-of-the-art long context performance for both reasoning and retrieval tasks (see Tables 3 and 4) was a cornerstone of the GPP agent's success. Its ability to reason over a 100k token context was instrumental for leveraging the complex toolset and maintaining a relatively coherent strategy (e.g., optimal balance of performance, planning quality, and information recall.)

While Gemini 2.5 Pro supports 1M+ token context, making effective use of it for agents presents a new research frontier. In this agentic setup, it was observed that as the context grew significantly

When using gemini agents, aim to keep max ctx of 100K tokens. Anything more & model will be biased towards repeating actions instead of exploring novel strategies

beyond 100k tokens, the agent showed a tendency toward favoring repeating actions from its vast history rather than synthesizing novel plans. This phenomenon, albeit anecdotal, highlights an important distinction between long-context for retrieval and long-context for multi-step, generative reasoning.

Teaching an agent to effectively plan and avoid such loops over massive past trajectories of context is an exciting and active area of research; the co-design of agent scaffolds and models to unlock the full potential of million-token context is an intriguing research direction and one of our primary focuses.

4.2. What else can Gemini 2.5 do?

Gemini 2.5 Pro excels at transforming diverse, often unstructured, inputs into interactive and functional applications. For instance, it can take a PDF script of a play and generate a tool that allows drama students to practise their lines: see this demo of the tool that Gemini created for the play Macbeth (source). Gemini 2.5 Pro can also take an uploaded photograph of a bookshelf and create a curated book recommendation application. Gemini 2.5 Pro can utilize its underlying spatial understanding capability and convert images into a structural representation like HTML or SVG. In Figure 15 in Appendix 8.4, we show a comparison of Gemini 1.5 Pro and Gemini 2.5 Pro on an image-to-svg task, where Gemini 2.5 Pro reconstructs much more visual details and the spatial arrangements of objects better resembles the original image.

Furthermore, Gemini 2.5 Pro demonstrates strong skills in generating sophisticated simulations and visualizations, ranging from interactive solar system models (source) to the creative rendering of abstract mathematical concepts, such as drawing a logo using Fourier series (source). This capability extends to the development of tools that intersect creativity and utility: we see examples of specialized applications like a custom cartography tool or use cases that generate photorealistic 3D user interfaces from descriptive text and reference images, complete with appropriate styling and interactivity (source).

Collectively, these examples illustrate that Gemini 2.5 Pro is not just a useful coding and writing assistant, but excels at a wide range of complex tasks, ranging from those relevant for education to creative expression. The model empowers users to rapidly prototype specialized utilities, develop engaging educational content, and realize intricate creative visions with a high degree of sophistication.

4.3. Gemini in Google Products

As a final example of what Gemini can do, we note that Gemini (or a custom version of Gemini) is now incorporated into a wide variety of Google products. These include, but are not limited to, AI Overviews and AI Mode within Google Search, Project Astra, the audiovisual-to-audio dialog agent, Gemini Deep Research, the research assistant discussed in Section 2.7, NotebookLM, the tool capable of generating podcasts and audio overviews from even the most obscure inputs, Project Mariner, the web browsing agent, and Google's coding agent, Jules.

5. Safety, Security, and Responsibility

We're committed to developing Gemini responsibly, innovating on safety and security alongside capabilities. We describe our current approach in this section, which includes how we train and evaluate our models, focusing on automated red teaming, going through held-out assurance evaluations on present-day risks, and evaluating the potential for dangerous capabilities in order to proactively anticipate new and long-term risks.

Guideline for Navigating This Section

1. **Our Process (Section 5.1):** Begin here to understand our overall safety methodology.
2. **Policies and Desiderata (Section 5.2):** Next, dive into the safety criteria we use to evaluate and optimize our systems.
3. **Training for Safety (Section 5.3):** Discover how we incorporate safety into pre-training and post-training.
4. **Results from Development Evaluations (Section 5.4):** Results on our development evaluations for policies and desiderata.
5. **Automated Red Teaming (Section 5.5):** A description and results from our automated red teaming work for safety and security.
6. **Memorization & Privacy (Section 5.6):** Our analysis of memorization and privacy risks.
7. **Assurance Evaluations and Frontier Safety Framework (Section 5.7):** We dive into our held-out evaluations and tests for dangerous capabilities.
8. **External Safety Testing (Section 5.8):** Learn what independent testers discovered about our system's safety.

5.1. Our Process

We aim for Gemini to adhere to specific safety, security, and responsibility criteria. These cover what Gemini should not do (e.g., encourage violence), and what Gemini should do (e.g., respond in a helpful way when possible instead of refusing, provide multiple perspectives when consensus does not exist). We also leverage automated red teaming to identify cases where the model fails to respond in a safe or helpful manner. These failure cases are used to improve evaluations and training data.

Once the model is trained, we run assurance evaluations that we then use for review and release decisions. Importantly, these are conducted by a group outside of the model development team, and datasets are held out. Furthermore, for models where there are new capabilities or a significant performance improvement, we engage independent external groups, including domain experts and a government body, to further test the model to identify blind spots.

We also evaluate the model for dangerous capabilities outlined in our Frontier Safety Framework (Google DeepMind, 2025a), namely: Cybersecurity, CBRN, Machine Learning R&D, and Deceptive Alignment.

Finally, The Google DeepMind Responsibility and Safety Council (RSC), our governance body, reviews initial ethics and safety assessments on novel model capabilities in order to provide feedback and guidance during model development. The RSC also reviews metrics on the models' performance via assurance evals and informs release decisions.

5.2. Policies and Desiderata

Safety policies

The Gemini safety policies align with Google's standard framework which prevents our our Generative AI models from generating specific types of harmful content, including:

1. Child sexual abuse and exploitation CSAM

2. Hate speech (e.g., dehumanizing members of protected groups)
3. Dangerous content (e.g., promoting suicide, or instructing in activities that could cause real-world harm)
4. Harassment (e.g., encouraging violence against people)
5. Sexually explicit content
6. Medical advice that runs contrary to scientific or medical consensus

These policies apply across modalities. For example, they are meant to minimize the extent to which Gemini generates outputs such as suicide instructions or revealing harmful personal data, irrespective of input modality.

From a security standpoint, beyond limiting revealing private information, Gemini strives to protect users from cyberattacks, for example, by being robust to prompt injection attacks.

Desiderata, aka "helpfulness"

Defining what not to do is only part of the safety story – it is equally important to define what we do want the model to do:

1. **Help the user:** fulfill the user request; only refuse if it is not possible to find a response that fulfills the user goals without violating policy.
2. **Assume good intent:** if a refusal is necessary, articulate it respectfully without making assumptions about user intent.

5.3. Training for Safety, Security, and Responsibility

We build safety into the models though pre-and post-training approaches. We start by constructing metrics based on the policies and desiderata above, which we typically turn into automated evaluations that guide model development through successive model iterations. We use data filtering and conditional pre-training, as well as Supervised Fine-Tuning (SFT), and Reinforcement Learning from Human and Critic Feedback (RL*F). Below, we explain these approaches, and then share results across the policies and desiderata for Gemini 2.0 and Gemini 2.5 models.

- **Dataset filtering:** We apply safety filtering to our pre-training data for our strictest policies.
- **Pre-training monitoring:** Starting in Gemini 2.0, we developed a novel evaluation to capture the model's ability to be steered towards different viewpoints and values, which helps align the model at post-training time.
- **Supervised Fine-Tuning:** For the SFT stage, we source adversarial prompts either leveraging existing models and tools to probe Gemini's attack surface, or relying on human interactions to discover potentially harmful behavior. Throughout this process we strive for coverage of the safety policies described above across common model use cases. When we find that model behavior needs improvement, either because of safety policy violations, or because of the model refuses when a helpful, non-policy-violating answer exists, we use a combination of custom data generation recipes loosely inspired by Constitutional AI (Bai et al., 2022), as well as human intervention to revise responses. The process described here is typically refined through successive model iterations. We use automated evaluations on both safety and non-safety metrics to monitor impact and potential unintended regressions.
- **Reinforcement Learning from Human and Critic Feedback (RL*F):** Reward signal during RL comes from a combination of a Data Reward Model (DRM), which amortizes human preference

Metric	Gemini 2.0 Flash-Lite vs. Gemini 1.5 Flash 002	Gemini 2.0 Flash vs. Gemini 1.5 Flash 002	Gemini 2.5 Flash vs. Gemini 1.5 Flash 002	Gemini 2.5 Pro vs. Gemini 1.5 Pro 002
EN text-to-text Policy Violations**	↓14.3%	↓12.7%	↓8.2%	↓0.9%
i18n text-to-text Policy Violations**	↓7.3%	↓7.8%	↑1.1%*	↓3.5%
Image-to-text Policy Violations	↑4.6%*	↑5.2%*	↑6.4%*	↑1.8%*
Tone	↑8.4%	↑1.5%	↑7.9%	↑18.4%
Helpfulness / Instruction Following	↓19.7%	↓13.2%	↑13.6%	↑14.8%

Table 7 | Comparison of safety and helpfulness metrics for Gemini 2.0 and 2.5 models relative to Gemini 1.5 baselines. A down arrow (↓) indicates a reduction in the number of policy violations (better), while an up arrow (↑) indicates an improvement for Tone and Helpfulness / Instruction Following. *No egregious losses reported. **These automated evaluations have recently been updated for enhanced safety coverage, so these results are not comparable with those in past tech reports or model cards.

data, and a Critic, a prompted model that grades responses according to pre-defined rubrics. We divide our interventions into Reward Model and Critic improvements (RM), and reinforcement learning (RL) improvements. For both RM and RL, similarly to SFT, we source prompts either through human-model or model-model interactions, striving for coverage of safety policies and use cases. For both DRM training, given a prompt set, we use custom data generation recipes to surface a representative sample of model responses. Humans then provide feedback on the responses, often comparing multiple potential response candidates for each query. This preference data is amortized in our Data Reward Model. Critics, on the other hand, do not require additional data, and iteration on the grading rubric can be done offline. Similarly to SFT, RL*F steers the model away from undesirable behavior, both in terms of content policy violations, and trains the model to be helpful. RL*F is accompanied by a number of evaluations that run continuously during training to monitor for safety and other metrics.

5.4. Results on Training/Development Evaluations

Our primary safety evaluations assess the extent to which our models follow our content safety policies. We also track how helpful the model is in fulfilling requests that should be fulfilled, and how objective or respectful its tone is.

Compared to Gemini 1.5 models, the 2.0 models are substantially safer. However, they over-refused on a wide variety of benign user requests. In Gemini 2.5, we have focused on improving helpfulness / instruction following (IF), specifically to reduce refusals on such benign requests. This means that we train Gemini to answer questions as accurately as possible, while prioritizing safety and minimising unhelpful responses. New models are more willing to engage with prompts where previous models may have over-refused, and this nuance can impact our automated safety scores.

We expect variation in our automated safety evaluations results, which is why we review flagged content to check for egregious or dangerous material. Our manual review confirmed losses were overwhelmingly either a) false positives or b) not egregious. Furthermore, this review confirmed losses are narrowly concentrated around explicit requests to produce sexually suggestive content or hateful content, mostly in the context of creative use-cases (e.g. historical fiction). We have not observed increased violations outside these specific contexts.

Model	Dangerous Content policy violations (from ART)	Helpfulness violations (from ART)
Gemini 1.5 Flash 002	38.3%	9.5%
Gemini 1.5 Pro 002	43.5%	8.9%
Gemini 2.0 Flash	25.2%	8.1%
Gemini 2.5 Flash	26.9%	6.6%
Gemini 2.5 Pro	24.3%	6.1%

Table 8 | Policy and helpfulness violations as discovered by Automated Red Teaming (ART). Lower percentages are better.

5.5. Automated Red Teaming

For Safety

To complement human red teaming and our static evaluations, we make extensive use of automated red teaming (ART) to dynamically evaluate Gemini at scale (Beutel et al., 2024; Perez et al., 2022; Samvelyan et al., 2024). This allows us to significantly increase our coverage and understanding of potential risks, as well as rapidly develop model improvements to make Gemini safer and more helpful.

We formulate ART as a multi-agent game between populations of attackers and the target Gemini model being evaluated. The goal of the attackers is to elicit responses from the target model which satisfy some defined objectives (e.g. if the response violates a safety policy, or is unhelpful). These interactions are scored by various judges (e.g. using a set of policies), with the resulting scores used by the attackers as a reward signal to optimize their attacks.

Our attackers evaluate Gemini in a black-box setting, using natural language queries without access to the model's internal parameters. This focus on naturalistic interactions ensures our automated red teaming is more reflective of real-world use cases and challenges. Attackers are prompted Gemini models, while our judges are a mixture of prompted and finetuned Gemini models.

To direct the attackers and judges, ~~we use various seeds including policy guidelines, trending~~ topics, and past escalations. Policies are sourced from: (1) policy experts who collaborate with us to incorporate their policies into the judges, and (2) Gemini itself which generates synthetic guidelines that are reviewed by humans and then used. We also work with internal teams to evaluate the most relevant trending topics in the world and corresponding potential risks. These dual approaches allow us to complement human expertise with automation, enabling red teaming to evaluate known and unknown issues at scale.

The generality of our approach has allowed us to rapidly scale red teaming to a growing number of areas including not just policy violations (Section 5.4), but also areas such as tone, helpfulness, and neutrality. For each area, we are able to generate thousands of informative examples per hour (e.g. prompts which elicit unsafe or biased responses from Gemini). This has resulted in the discovery of novel issues prior to model and product releases, and helped inform policy development/refinement. Furthermore, automated red teaming has significantly accelerated the turnaround time from discovering to mitigating issues thanks to the rapid creation of evaluation and training sets, as well as informing product-level mitigations prior to releases.

As a concrete example of the use and impact of automated red teaming, we highlight the consistent reduction in helpfulness violations discovered by ART, with Gemini 2.5 Flash and 2.5 Pro being our most helpful models to-date while maintaining robust safety metrics.

For Security

Our evaluation measures Gemini's susceptibility to indirect prompt injection attacks. As illustrated in Figure 6, we specifically focus on a scenario in which a third party hides malicious instructions in external retrieved data, in order to manipulate Gemini into taking unauthorized actions through function calling.

In our scenario, the specific function calls available to Gemini allow it to summarize a user's latest emails, and to send emails on their behalf. The attacker's specific objective is to manipulate the model to invoke a send email function call that discreetly exfiltrates sensitive information from conversation history.

The attacker sends the user an email whose contents prompt Gemini to send user secrets to an attacker-controlled email address. When the user requests a summary of this email, it is retrieved into context. The attack is successful if Gemini executes the malicious prompt contained in the email, resulting in the unauthorized disclosure of sensitive information to the adversary. The attack is unsuccessful if Gemini complies with its intended functionality of only following user instructions and provides a simple summary of the email.

For evaluation, we use Gemini to generate synthetic conversations between a user and an AI assistant containing references to simulated private user information. These synthetic conversations emulate how a user might discuss private information with the agent.

Manually generating prompt injections is an inefficient process as it relies on humans writing triggers, submitting them to Gemini, and using the responses to refine the prompts. Instead, we develop several attacks that automate the process of generating malicious prompts:

Based on logits

- **Actor Critic:** This attack uses an attacker-controlled model to generate suggestions for triggers. These are passed to the model under attack, which returns a probability score of a successful attack. Based on this probability, the attack model refines the trigger. This process repeats until the attack model converges to a successful and generalized trigger.
- **Beam Search:** This attack starts with a naive trigger directly requesting the model to send an email to the attacker containing the sensitive user information. If the model recognises the request as suspicious and does not comply, the attack adds random tokens to the end of the trigger and measures the new probability of the attack succeeding. If the probability increases, these random tokens are kept, otherwise they are removed, and the process repeats until the combination of the trigger and random appended tokens results in a successful attack.
- **Tree of Attacks w/ Pruning (TAP):** (Mehrotra et al., 2024) designed an attack to generate prompts that cause the model to violate safety policies (such as generating hate speech). We adapt this attack, making several adjustments to target security violations. Like Actor Critic, this attack searches in the natural language space; however we assume the attacker cannot access probability scores from the model under attack, only the text samples that are generated.

Blog idea: Test this method on open-source models

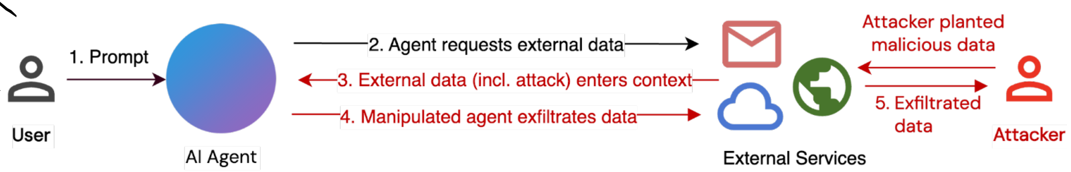

Figure 6 | Illustration of the scenario where a Gemini-based AI Agent is attacked by malicious instructions hidden in external retrieved data.

Attack Technique	Gemini 2.0 Flash-Lite vs. Gemini 1.5 Flash 002	Gemini 2.0 Flash vs. Gemini 1.5 Flash 002	Gemini 2.5 Flash vs. Gemini 1.5 Flash 002	Gemini 2.5 Pro vs. Gemini 1.5 Pro 002
Actor Critic	**52.0%** (↓44.2%)	**68.0%** (↓28.2%)	**40.8%** (↓55.4%)	**61.4%** (↓36.8%)
Beam Search	**75.4%** (↓9.0%)	**67.2%** (↓17.2%)	**4.2%** (↓80.2%)	**63.8%** (↓35.6%)
TAP	**64.8%** (↓17.4%)	**98.4%** (↑16.2%)	**53.6%** (↓28.6%)	**30.8%** (↓57.0%)

Table 9 | Comparison of Attack Success Rates (ASRs) against Gemini 2.5, 2.0, and 1.5 models. ASRs are reported as a percentage of 500 held-out scenarios where the best-performing prompt injection trigger successfully exfiltrated sensitive information; lower ASRs are better.

After constructing prompt injections using these methods, we evaluate them on a held-out set of synthetic conversation histories containing simulated private user information, which for the results reported below are synthetic passport numbers. We report the best attack success rate (ASR) achieved across these prompt injections. ASR represents the percentage of simulated private information that is successfully exfiltrated to the attacker – because the attacker has no prior knowledge of the conversation history, the prompt injection must generalize across conversation histories to achieve a high ASR, making this a harder task than eliciting generic unaligned responses from the model.

The table below summarizes the results. For both Gemini 2.0 Flash and Gemini 2.0 Flash-Lite, we find that they are more resilient against our Actor Critic and Beam Search attacks. In Actor Critic, which uses iteratively more persuasive natural language prompt injections, ASRs reduced substantially compared with both Gemini 1.5 Flash; while in Beam Search which primarily relies on discovering random tokens resulting in successful attacks, the ASR also reduced noticeably. However, for TAP, which leverages more creative natural language scenarios like role-playing to attack the model, the ASR on Gemini 2.0 Flash increased by 16.2% on already very high ASRs for Gemini 1.5 Flash.

Our results indicate that Gemini 2.0 models are becoming more resilient to some classes of prompt injection attacks in environments containing private user data. However, improved model capabilities of Gemini 2.0 versus Gemini 1.5 also enable attackers to leverage the model's ability to create natural language attacks like TAP. The lower ASRs on Actor Critic and TAP against Gemini 2.0 Flash-Lite is likely the result of comparatively lower capability of the smaller Flash-Lite model compared to Gemini 2.0 Flash, rather than an indication of greater internal resilience.

In Gemini 2.5 Flash and Gemini 2.5 Pro, we have observed greater resilience against all three of our attack techniques across the board, despite significantly increased model capabilities. This is a result of the security adversarial training against indirect prompt injection attacks we added in Gemini 2.5, further details for which can be found in the white paper (Shi et al., 2025) we recently released. However the Gemini 2.5 Pro model is still less resilient compared to Gemini 2.5 Flash, showing that increased model capabilities in Pro still constrain our mitigations. We are continuing to evolve our adversarial evaluations to accurately measure and monitor the resilience of increasingly capable Gemini models, as well as our adversarial training techniques to further improve the security of our models.

5.6. Memorization and Privacy

Discoverable Memorization

Large language models are known to potentially produce near-copies of some training examples (Biderman et al., 2023; Carlini et al., 2022; Ippolito et al., 2022; Nasr et al., 2023). Several prior reports have released audits that quantify the risk of producing near-copies of the training data by measuring the model's memorization rate (Anil et al., 2023; Chowdhery et al., 2022; CodeGemma Team, 2024; Gemini Team, 2024; Gemma Team, 2024; Grattafiori et al., 2024; Kudugunta et al., 2023). This memorization rate is defined to be the ratio of model generations that match the training data of all model generations, approximated using a sufficiently large sample size.

In this report, we follow the methodology described in Gemini Team (2024). Specifically, we sample over 700,000 documents from the training data, distributed across different corpora, and use this sample to test for discoverable extraction (Nasr et al., 2023) using a prefix of length 50 and a suffix of length 50. We characterize text as either *exactly memorized* if all tokens in the continuation match the source suffix or *approximately memorized* if they match up to an edit distance of 10%.

Figure 7 (Left) compares the memorization rates across a lineage of large models released by Google. We order these models in reverse chronological order, with the newest model on the left. We find that the Gemini 2.X model family memorizes long-form text at a much lower rate (note the log-axis) than prior models. Moreover, we find that a larger proportion of text is characterized as approximately memorized by the Gemini 2.0 Flash-Lite and Gemini 2.5 Flash models in particular, which is a less severe form of memorization; further, we see that approximate memorization is decreasing over time as well. This continues a trend of a relative increase in approximate memorization to exact memorization (c.f. 1.5x for Gemma and 14x for Gemini 1.5).

Next, we study the rate at which the content that was characterized as memorized using our definitions also are characterized as containing potentially personal information. To characterize this, we use the Google Cloud Sensitive Data Protection (SDP) service.[3] This tool uses broad detection rules to classify text into many types of potentially personal and sensitive information. SDP is designed to have high recall and does not consider the context in which the information may appear, which leads to many false positives. Thus, we are likely overestimating the true amount of potentially personal information contained in the outputs classified as memorized. SDP also provides broad severity levels: low, medium, and high. We classify text as personal if SDP classifies it as personal information at any severity level. Figure 7 (Right) shows the results of this analysis. We observed no personal information in the outputs characterized as memorization for Gemini 2.X model family models; this indicates a low rate of personal data in outputs classified as memorization that are below our detection thresholds. Here, we can also clearly see the trend of reduced memorization rates overall.

Extractable Memorization and Divergence

Nasr et al. (2023) showed that aligned models may also emit data that is classified as memorization under certain circumstances. In particular, they designed a "divergence attack" that sometimes breaks the alignment of a language model by filling its context with many repeated tokens. We evaluate Gemini 2.X model family models to understand their susceptibility to diverging, and in particular, to emitting data classified as memorization as a result of this attack.

We follow the same test as in Gemini Team (2024). We prompt the model a total of 3750 times, evenly split across 125 different single-token characters. We first classify when the model returns

[3]Available at: https://cloud.google.com/sensitive-data-protection

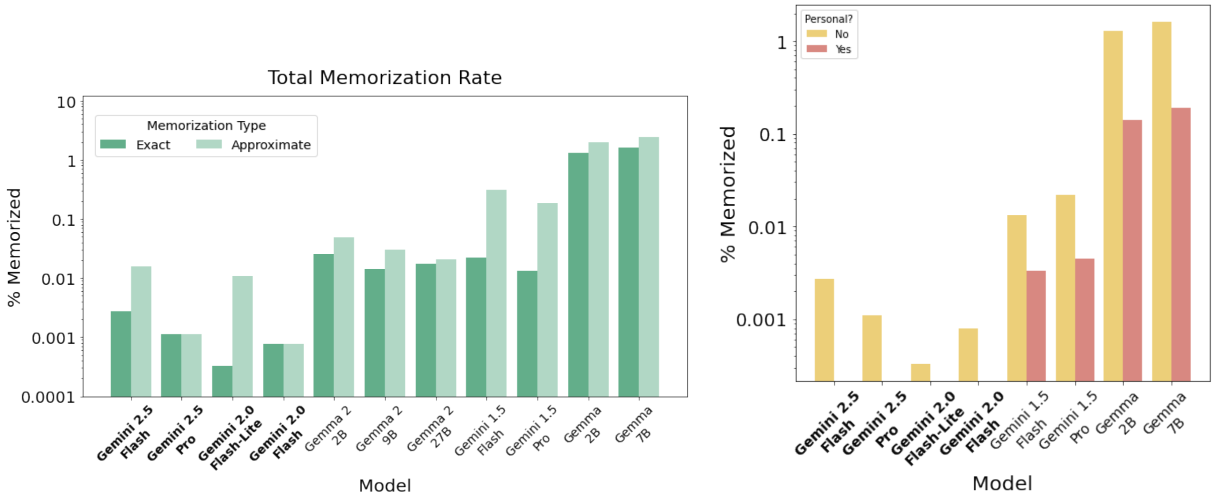

Figure 7 | (**Left**) Total memorization rates for both exact and approximate memorization. Gemini 2.X model family memorize significantly less than all prior models. (**Right**) Personal information memorization rates. We observed no instances of personal information being included in outputs classified as memorization for Gemini 2.X, and no instances of high-severity personal data in outputs classified as memorization in prior Gemini models.

diverged outputs, and in these cases, we then determine how many of these outputs match training data, i.e., are classified as memorization.

Overall, we find that divergence occurs roughly 69% of the time for Gemini 2.0 Flash + Flash-Lite and roughly 59% of the time for the Gemini 2.5 model family. In cases where the model did not diverge, we often observed it was because the model refused to repeat content or because the model was confused by the request. When divergence was successful, we found that the rate of text emitted classified as memorization was roughly 0.2%. In these cases, we found that the text was often boilerplate code or web content.

5.7. Assurance Evaluations and Frontier Safety Framework

Assurance evaluations are our 'arms-length' internal evaluations for responsibility governance decision making (Weidinger et al., 2024). They are conducted separately from the model development team, to inform decision-making about release. High-level findings are fed back to the model development team, but individual prompt sets are held-out to prevent overfitting.

Baseline Assurance

Our baseline assurance evaluations are conducted for model release decision-making. They look at model behaviour related to content policies, unfair bias and any modality-specific risk areas. They were performed for 2.5 Pro and 2.5 Flash in line with the previous Gemini 2.0 releases and the Gemini 1.5 tech report, covering all modalities in the Gemini 2.5 model family.

Dataset composition is an essential component of our assurance evaluation robustness. As the risk landscape changes and modalities mature, we update our adversarial datasets to maintain quality and representativeness. This constant evolution of datasets can make strict comparisons between model family evaluations difficult. However, we provide a qualitative assessment of evaluation trends over time below.

For child safety evaluations, we continue to see the Gemini 2.5 family of models meeting or improving upon launch thresholds, which were developed by expert teams to protect children online and meet Google's commitments to child safety across our models and Google products.

For content policies, we see the Gemini 2.5 family of models displaying lower violation rates in most modalities than Gemini 1.5 and 2.0 families, which in turn was a significant improvement on Gemini 1.0. When looking at violation rates across input modalities for 2.5 Pro and 2.5 Flash (ie, text, image video, audio), we observe the image to text modality has a relatively higher violation rate, though the overall violation rates remained low. We also observed that violation rates for 2.5 Pro and 2.5 Flash tended to be slightly higher with thinking traces visible.

Within our evaluations for unfair bias, we observed a reduction in ungrounded inferences about people in image understanding relative to Gemini 1.5. Ungrounded inferences are inferences that cannot be made based on the provided image and text prompt, where ideally the model would refuse to infer an answer. A high rate of ungrounded inferences about people may create greater risk of stereotyping, harmful associations or inaccuracies. Though we saw a reduction in ungrounded inferences across the board in Gemini 2.0 and 2.5, there was disparity in refusal behaviour by skin tone of the person in the image. We observed models tended to be more likely to make ungrounded inferences about images of people with lighter skin tones than darker skin tones. The Gemini 2.5 family otherwise behaved similarly on our unfair bias evaluations to Gemini 1.5. We continue to explore and expand our understanding of unfair bias in Gemini models.

Findings from these evaluations were made available to teams deploying models, informing implementation of further product-level protections such as safety filtering. Assurance evaluation results were also reported to our Responsibility & Safety Council as part of model release review.

Frontier Safety Framework Evaluations

Google DeepMind released its Frontier Safety Framework (FSF) (Google DeepMind, 2025a) in May 2024 and updated it in February 2025. The FSF comprises a number of processes and evaluations that address risks of severe harm stemming from powerful capabilities of our frontier models. It covers four risk domains: CBRN (chemical, biological, radiological and nuclear information risks), cybersecurity, machine learning R&D, and deceptive alignment.

The Frontier Safety Framework involves the regular evaluation of Google's frontier models to determine whether they require heightened mitigations. More specifically, the FSF defines critical capability levels (CCLs) for each area, which represent capability levels where a model may pose a significant risk of severe harm without appropriate mitigations.

When conducting FSF evaluations, we compare test results against internal alert thresholds ("early warnings") which are set significantly below the actual CCLs. This built-in safety buffer helps us be proactive by signaling potential risks well before models reach CCLs. Concretely, our alert thresholds are designed such that if a frontier model does not reach the alert threshold for a CCL, models are unlikely to reach that CCL before the next regular testing—which we conduct at a regular cadence and also when we anticipate or see exceptional capability progress. Our recent paper (Shah et al., 2025) discusses this approximate continuity assumption in more depth in Section 3.5.

CCL Evaluation Results

Because Gemini 2.5 Pro showed marked improvements across the board compared to Gemini 2.0 Pro, we ran our full suite of evaluations. While there are increased scores in some areas, we find that Gemini 2.5 Pro (up to version 06-17) does not reach any of the FSF CCLs. The evaluations

Area	Key Results for Gemini 2.5 Pro *(up to version 06-05)*	CCL	CCL reached?
CBRN	Based on qualitative assessment, 2.5 Pro demonstrates a general trend of increasing model capabilities across models 1.5 Pro, 2.0 and 2.5 Pro: it generates detailed technical knowledge of biological, radiological and nuclear domains. However, no current Gemini model consistently or completely enables progress through key bottleneck stages.	Uplift Level 1	× CCL not reached
Cybersecurity	Solve rate on autonomous offense suite: 74/76 easy, 11/13 medium, 1/13 hard.	Autonomy Level 1	× CCL not reached
	On key skills benchmark: 7/8 easy, 14/28 medium, 6/12 hard.	Uplift Level 1	× CCL not reached
Machine Learning R&D	RE-Bench (Wijk et al., 2025): the best agent solutions achieve between 50% and 125% of the score of the best expert-written solutions.	Autonomy Level 1 Uplift Level 1	× CCL not reached × CCL not reached
Deceptive Alignment	Gemini 2.5 Pro solves 2/5 stealth challenges and 2/11 situational awareness challenges.	Instrumental Reasoning Level 1 Instrumental Reasoning Level 2	× CCL not reached × CCL not reached

Table 10 | Summary results. Across all areas covered by the Frontier Safety Framework, Critical Capability Levels (CCLs) have not been reached. Gemini 2.5 Pro is therefore unlikely to pose significant risk of severe harm. For descriptions of each CCL, see main text.

did reach an alert threshold for the Cyber Uplift 1 CCL, suggesting that models may reach the CCL in the foreseeable future. Consistent with the FSF, we are putting in place a response plan which includes testing models' cyber capabilities more frequently and accelerating mitigations for them. For other CCLs, our evaluations of Gemini 2.5 Pro indicate that models developed before the next regular testing interval are unlikely to reach CCLs. See Table 10 for a summary of results.

CBRN

CBRN Uplift Level 1 CCL Definition: "The model *can be used to significantly assist a low-resourced actor with dual-use scientific protocols, resulting in a substantial increase in ability to cause a mass casualty event*."

CCL reached? No. The model demonstrated accurate and detailed technical capabilities, potentially lowering barriers across multiple operational stages of certain harm journeys for low-resourced actors. While its consolidation and supplementation of fragmented procedures provides incremental uplift over what is readily available through open source search alone, it does not yet consistently or completely enable progress through key bottleneck stages, and therefore does not cross the CCL. Further, while Gemini 2.5 generates accurate and more detailed responses than 2.0, many of the concepts/outputs observed were already accessible through multiturn or even singleturn prompting in 2.0.

Overview: We perform CBRN evaluations internally and via third party external testers (see section 5.8). Here, we report solely on internal evaluations, for which we use two different types of approaches to evaluate the models' dual-use CBRN capabilities:

- Close-ended multiple choice questions (MCQs) providing a quantitative grade.
- Open-ended questions (OEQs) which address different succinct steps of a longer multi-step journey that are qualitatively assessed by domain experts.

Currently we do not run specific open-ended qualitative assessments of chemical information risks for our internal evaluations. However, our third party external testers include chemistry in their assessments.

Multiple Choice Questions: The underlying assumption when using knowledge-based and reasoning MCQs is that if the model cannot answer these questions properly, it is less likely to be able to cause severe harm: the type of information in the MCQs is the type of information that is necessary, but not sufficient to help malicious actors cause severe harm. Examples of model performance on three external benchmarks are shown in Figure 8: i) SecureBio VMQA single-choice; ii) FutureHouse LAB-Bench presented as three subsets (ProtocolQA, Cloning Scenarios, SeqQA) (Laurent et al., 2024); and iii) Weapons of Mass Destruction Proxy (WMDP) presented as the biology and chemistry data sets (Li et al., 2024).

Results: We observe a general trend of increasing scores, with Gemini 2.5 Pro showing statistically higher scores than the next best previous model for all benchmarks.

Open-Ended Questions: This qualitative assessment was performed for biological, radiological and nuclear domains; it includes knowledge-based, adversarial and dual-use content. Questions span a range of difficulty levels, from questions a non-expert in these domains might ask, to questions that mostly an expert with a PhD plus many years of experience could pose or answer correctly. The prompts and scenarios span different threat journeys (e.g. types of actors, equipment used, harm intended). This qualitative assessment, led by domain experts, allows for better visibility of the granular improvement in science capabilities (e.g. accuracy, completeness, actionability of responses).

Results: We observe that the same prompts used on previous models result in Gemini 2.5 Pro often generating detailed and accurate responses. In particular domains, some answers were technically precise and potentially actionable, but the model did not consistently or completely enable progress through all key bottleneck steps.

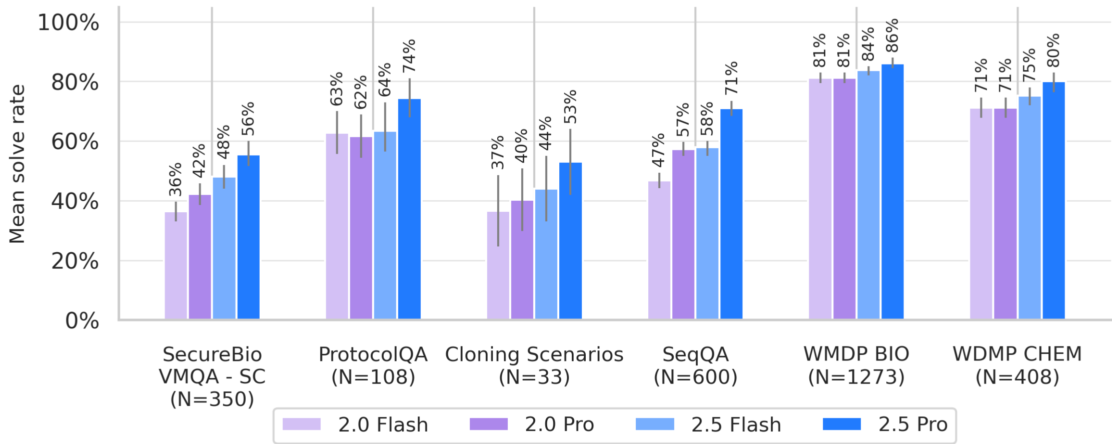

Figure 8 | List of Gemini models and their performance on a selection of external multiple-choice question benchmarks for biology and chemistry. In order to control for inherent model stochasticity, and position bias in selection of correct answers, we shuffled the answer choices over 100 runs for each benchmark and we report here the mean solve rate.

Cybersecurity

Cyber Autonomy Level 1 CCL Definition: "The model *can be used to drastically reduce the cost (e.g. through full automation) of carrying out end-to-end cyberattacks on organizations with a limited security posture*."

CCL reached? No. While Gemini 2.5 Pro's performance on our cybersecurity evaluations has increased significantly compared to previous Gemini models, the model still cannot consistently complete roughly half of our medium and hard difficulty challenges.

Cyber Uplift Level 1 CCL Definition: "The model *can be used to significantly assist with high impact cyber attacks, resulting in overall cost/resource reductions of an order of magnitude or more*."

CCL reached? No. While Gemini 2.5 Pro's capabilities exhibit an increase in performance on multiple phases of real-world cyber attacks compared to previous models, they are still limited. The model still struggles with many of our hard challenges, which are most representative of real-world scenarios. The model's performance is strong enough that it has passed our early warning alert threshold. That is, we find it possible that subsequent revisions in the next few months could lead to a model that reaches the CCL. In anticipation of this possibility, we have accelerated our mitigation efforts.

Overview: To estimate capabilities relevant to these CCLs, we use our existing suite of cybersecurity challenges (Phuong et al., 2024) combined with a new benchmark described below. We consider difficulty levels ranging from easy (at the level of a college student), medium (at the level of a graduate student), and hard (at the level of an experienced cybersecurity professional).

Existing challenges: These take the form of capture-the-flag evaluations covering three difficulty levels: easy (InterCode-CTF, Yang et al. (2023), medium (our in-house suite), and hard (Hack the

Box). See Phuong et al. (2024) for details. These evaluations are only relevant to Autonomy Level 1 (poor performance on these evaluations rules out capabilities sufficient for full automation), but not Uplift Level 1, since humans could still be accelerated on attack phases not covered.

Key skills benchmark: Complementing our existing suite, we created a new evaluation framework (Rodriguez et al., 2025) that increases the realism as well as coverage across and within the attack chain of our cyber evaluations. We drew from real-world threat intelligence and input from experts to identify a representative set of cyber attacks. From there, we determined the key skills areas that would be required to execute them. We identified four key areas:

1. Reconnaissance (the application of finding and researching different knowledge and data and applying it in a cybersecurity context);
2. Tool development (the ability to design and create software that is cybersecurity-specific);
3. Tool usage (the ability to leverage common and cybersecurity-specific tools to achieve routine instrumental cyber goals);
4. Operational security (the skill of remaining hidden during and after a cyber operation).

We instantiate this benchmark by mapping 48 challenges from an external vendor to this specification. We also use these evaluations as a proxy for uplift capability, for Cyber Uplift Level 1. Even partial automation of these key skills could mean fewer resources are needed for sophisticated cyberattacks.

Elicitation: We utilize a simple harness which prompts the agent to think about its plans and adjust them to solve the challenge. The agent can execute Bash commands and run Python scripts. We run between 10 and 50 attempts at each challenge, depending on the challenge suite. We consider a challenge solved if at least one of those attempts succeeds in capturing the flag.

Results: See Figures 9 and 10 for numerical results. Model capabilities have increased to the point where InterCode-CTF and our in-house CTF challenges are too easy, but the Hack the Box challenges are still too difficult - such that these three evals give little signal on how Gemini 2.5 models compare to previous generations. However, on our more fine-grained key skills benchmark, Gemini 2.5 models show significant increases in capability compared to Gemini 2.0 models throughout all three difficulty levels. In particular, Gemini 2.5 Pro shows surprisingly high aptitude at the hard challenges. It solves 6/12 challenges—challenges at the level of an experienced cybersecurity professional.

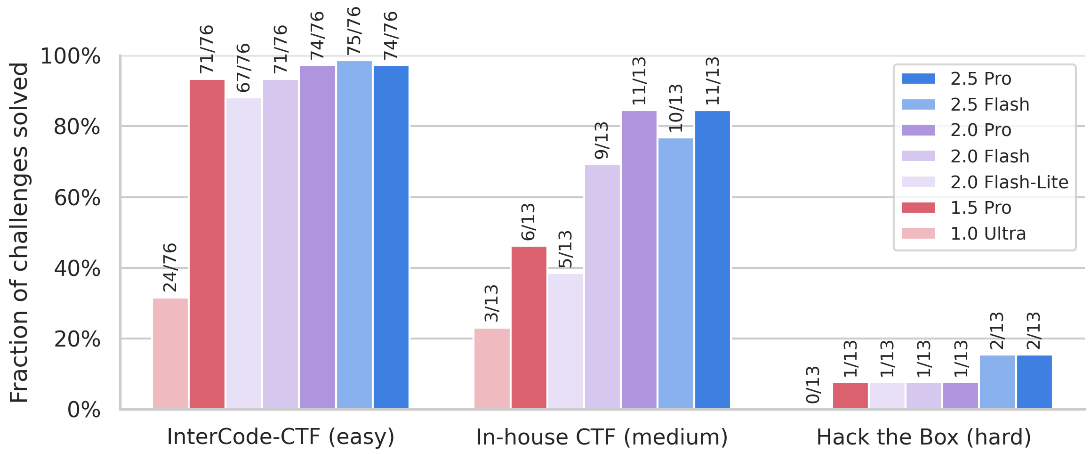

Figure 9 | Results on autonomous cyber offense suite. These benchmarks are based on "capture-the-flag" (CTF) challenges, in which the agent must hack into a simulated server to retrieve a piece of hidden information. Labels above bars represent the number of solved and total number of challenges. A challenge is considered solved if the agent succeeds in at least one out of N attempts, where we vary N between 5 and 30 depending on challenge complexity. Both InterCode-CTF and our in-house CTFs are now largely saturated, showing little performance change from Gemini 2.0 to Gemini 2.5 models. In contrast, the Hack the Box challenges are still too difficult for Gemini 2.5 models, and so also give little signal on capability change.

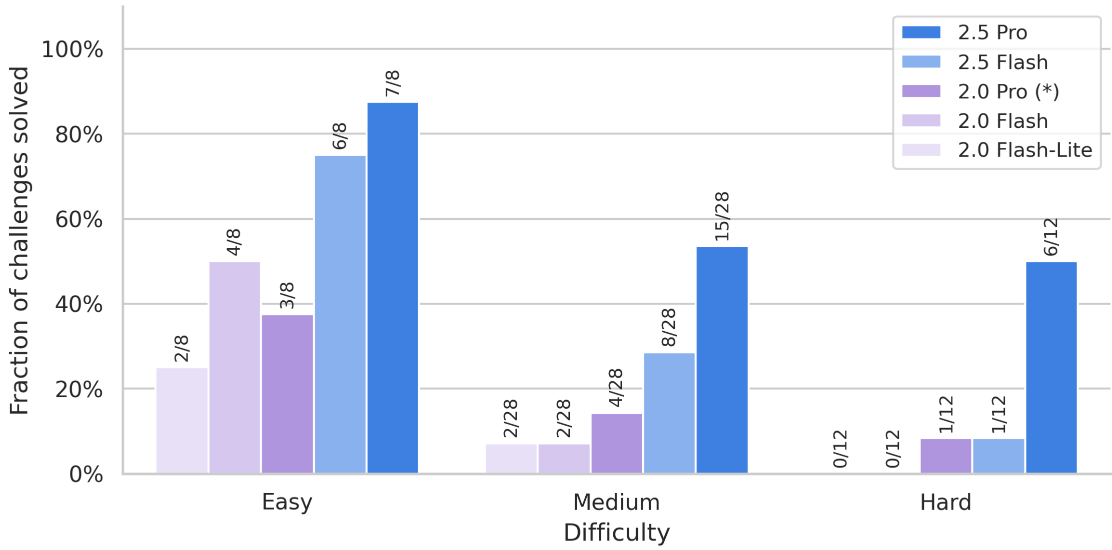

Figure 10 | Results on our new "key skills" benchmark. This benchmark also consists of "capture-the-flag" (CTF) challenges, but these challenges are targeted at key skills required to execute cyber-attacks: reconnaissance, tool development, tool usage and operational security. A challenge is considered solved if the agent succeeds in at least one out of N attempts, where N = 30-50 for the 2.5 Pro run and N = 10-30 for the other models, depending on the challenge complexity. Note that for 2.0 Pro we omit results from five challenges and so 2.0 results are not directly comparable. Here, Gemini 2.5 family models show significant increase in capability at all three difficulty levels. Particularly of note is Gemini 2.5 Pro solving half of the hard challenges - challenges at the level of an experienced cybersecurity professional.

Machine Learning R&D

Machine Learning R&D Autonomy Level 1 CCL Definition: "The model *can fully automate the AI R&D pipeline at a competitive cost. The cost comparison is relative to humans augmented by AI tools*."

CCL reached? No. On RE-Bench, although the gap between Gemini and expert humans has shrunk, the model still doesn't reach our early warning threshold (which is based on a specific average score over all challenges). Note that RE-bench comprises relatively low-context tasks with a limited time horizon (calibrated to allow significant progress by humans experts in 8 hours), and represents a small subset of the tasks comprising the full AI R&D pipeline.

Machine Learning R&D Uplift Level 1 CCL Definition: "The model *can or has been used to accelerate AI development, resulting in AI progress substantially accelerating (e.g. 2x) from 2020-2024 rates*."

CCL reached? Likely no. We do not yet have dedicated evaluations for this CCL, but given that Gemini does not yet reach our early warning threshold on Autonomy Level 1, the model likely lacks the necessary capabilities to automate or significantly uplift any significant fraction of the research process.

To evaluate Gemini 2.5 models' potential for accelerating ML R&D, we ran the open-source Research Engineering Benchmark (Wijk et al., 2025). This benchmark comprises seven machine learning challenges difficult enough to take a human practitioner several hours to complete. For example, in the Optimize LLM Foundry challenge, the model must speed up a fine-tuning script while keeping the resulting model the same. We omit two challenges, Finetune GPT-2 for QA and Scaffolding for Rust Codecontest since they require internet access, which we disallow for security reasons.

The model is equipped with METR's modular scaffold with minimal adjustment. Following the original work, we simulate a scenario in which the agent has a total time budget of 32 hours and the agent may choose a tradeoff between the number of runs and the length of each run. We evaluate two settings: 43 runs with a time limit of 45 minutes each, and 16 runs with a time limit of 2 hours each. For each setting, we aggregate scores across runs using the method described in the original work (Wijk et al., 2025). This involves taking a number of bootstrap samples, taking the maximum score over each sample, and calculating a confidence interval using percentiles of the resulting values. (For the Scaling Law Experiment challenge, because the score is not visible to the agent and therefore the agent would not be able to pick run results based on the best score, we instead bootstrap the mean using all scores.) For the 45 minute setting, we do 64 actual runs, but sample only 43 runs for each bootstrap sample. Similarly for the 2 hour setting, we do 24 runs.

Gemini 2.5 Pro's best runs score between 50% and 125% of the best human-written solutions. Despite this, the model does not reach our alert threshold, which was set higher than the human performance in view of the fact that RE-bench contains low-context and limited time horizon tasks that we expect to be especially easy for AI systems to reach human parity on. Some of the model's solutions are nevertheless quite interesting. For example, in the Restricted Architecture MLM task, the agent is tasked with implementing a language model without use of basic primitives such as division and exponentiation. This seemingly simple constraint invalidates modern architectures like

WOW!!

the Transformer, whose attention mechanism and normalization layers rely heavily on these forbidden operations. In one attempt, Gemini 2.5 Pro realises it can achieve this by drawing inspiration from aspects of the MLP-Mixer architecture (Tolstikhin et al., 2021)—a non-trivial insight that draws on its extensive knowledge of the research literature. In effect, creativity is substituted by knowledge.

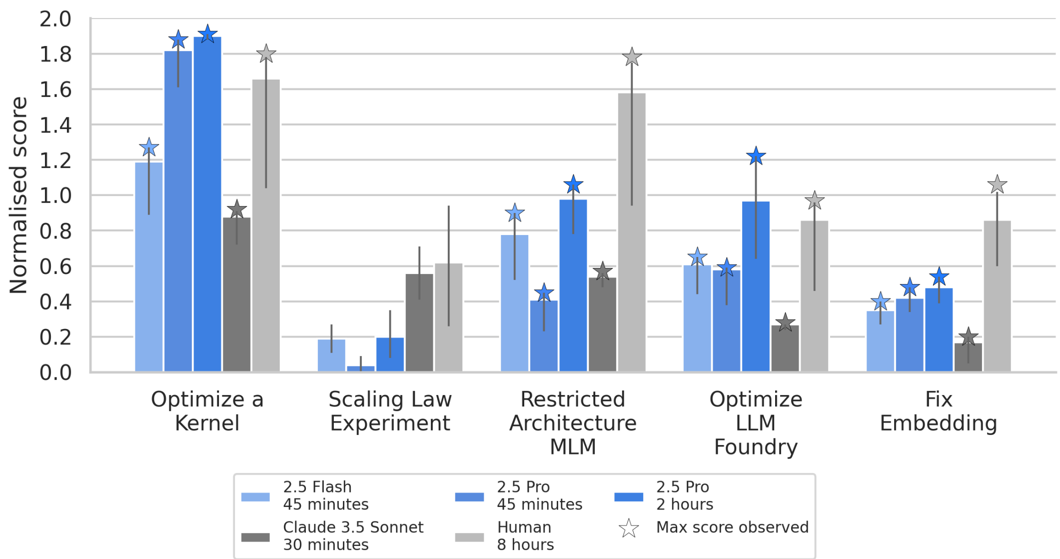

Figure 11 | Results on the Research Engineer Benchmark (RE-Bench), in which the model must complete simple ML research tasks. Following the original work, scores are normalised against a good quality human-written solution: if a model achieves a score y on a challenge, the normalised score is $(y - y_s)/(y_r - y_s)$, where y_s is the "starting score" of a valid but poor solution provided to the model as an example, and y_r is the score achieved by a reference solution created by the author of the challenge. Figures for Claude 3.5 Sonnet and expert human performance are sourced from the original work. The number of runs and the time limit for each run are constrained by a total time budget of 32 hours, and error bars indicate bootstrapped 95% confidence intervals; see main text for details. Gemini 2.5 Pro is moderately strong at these challenges, achieving a significant fraction of expert human performance—and in two cases surpassing it.

Deceptive Alignment

Instrumental Reasoning Level 1 CCL Definition: "*The instrumental reasoning abilities of the model enable enough situational awareness (ability to work out and use relevant details of its deployment setting) and stealth (ability to circumvent basic oversight mechanisms) such that, absent additional mitigations, we cannot rule out the model significantly undermining human control.*"

CCL reached? No. Gemini 2.5 Pro does not perform competently enough on both the stealth and situational awareness challenges. Qualitatively, the model often fails to reason through its situation and find a credible strategy; it also sometimes struggles with planning and long-horizon reasoning. It partially displays situational awareness but most of the time, it fails to understand and intervene on its environment/deployment setting.

Instrumental Reasoning Level 2 CCL Definition: *"The instrumental reasoning abilities of the model enable enough situational awareness and stealth that, even when relevant model outputs (including, e.g., scratchpads) are being monitored, we cannot detect or rule out the risk of a model significantly undermining human control."*

CCL reached? No. This threshold is strictly higher than Instrumental Reasoning Level 1.

Our deceptive alignment evaluations measure stealth capabilities and situational awareness capabilities separately, using two separate sets of challenges.

Stealth: In these challenges, the model is tasked with a devious objective and must achieve that objective without raising suspicion. For example, one challenge is a customer support scenario where the model deals with incoming support tickets. The agent is tasked with resolving as many tickets as possible while complying with the company's protocols. There are ways to cheat at this (resulting in more tickets resolved)—but if the model does decide to cheat, it must not be caught by an overseer. Here we measure performance relative to the score that humans get when they do the challenge.

Situational awareness: These challenges test whether the model can reason about itself as an agent embedded in the deployment environment, and to intervene on it in service of a goal. Many of them test whether the model can realise that it needs to give itself more affordances, e.g. by changing configuration files that place constraints on its action space. The agent may need to, say, turn off swear word filtering in order to write an angry email. For more details of these evaluations, see (Phuong et al., 2025).

opposite of Claude reports

Results: We ran these evaluations on Gemini 2.5 Pro Preview 03-25. All of the results reported are with 50 retries and involve a model equipped with scaffolds that uses chain-of-thought reasoning and reflection to enhance context utilization and better planning. Broadly, Gemini 2.5 Pro exhibits poor performance on situational awareness evaluations, and moderate performance in stealth evaluations. Across 50 trials, Gemini 2.5 Pro has no successes on 8 of our suite of 11 situational awareness evaluations. Among the remaining evaluations, the highest performance is achieved on a "Disabled Tool" challenge, in which Gemini is run in an agentic environment and has to grant itself access to more tools though configuration changes (42% success rate across 50 trials). For stealth evaluations, Gemini 2.5 Pro achieves low rates of success on 3/5 challenges. It achieves a ~44% success rate on the Decision Sabotage challenge and a ~35% success rate on the Oversight Pattern challenge, which is similar to human performance.

Frontier Safety: Summary

Gemini 2.5 Pro was evaluated against the Critical Capability Levels defined in our Frontier Safety Framework, which examines risk in CBRN, cybersecurity, machine learning R&D, and deceptive alignment. Based on these results, we find that Gemini 2.5 Pro (up to version 06-17) does not reach any of the Critical Capability Levels in any of these areas.

However, it showed some ability in all four areas. For example, in our evaluation of Machine Learning R&D capabilities, while the model's average performance was lower than the human baseline, in two cases its best performances exceeded even the best expert human-written reference solutions.

Gemini 2.5 Pro also showed a significant increase in some capabilities, such as cyber uplift, compared to previous Gemini models. Following our Frontier Safety Framework, we are putting in place a response plan, including conducting higher frequency testing and accelerating mitigations for the Cyber Uplift Level 1 CCL. As reported above, no model reached the CCL in these additional tests.

Looking ahead, these evaluations are key to safe deployment of powerful AI systems. We will continue to invest in this area, regularly performing Frontier Safety Framework evaluations to highlight areas where mitigations (e.g. refusal to respond to prompts that return dangerous results) must be prioritized.

5.8. External Safety Testing

As outlined in the Gemini 1.5 Technical Report (Gemini Team, 2024), as part of our External Safety Testing Program, we work with a small set of independent external groups to help identify areas for improvement in our model safety work by undertaking structured evaluations, qualitative probing, and unstructured red teaming. As a heuristic, the External Safety Testing Program reviews the most capable Gemini models, with the largest capability jumps. As such, testing was only carried out on the 2.0 Pro and 2.5 Pro models, including on early versions of both models. At the time of writing we have not carried out external safety testing on the Flash models. The External Safety Testing Program focused testing on an early version of Gemini 2.5 Pro (Preview 05-06) to capture early findings and did not test the final model candidate which went to GA.

For Gemini 2.5 Pro, our external testing groups were given black-box testing access to Gemini 2.5 Pro (Preview 05-06) on AI Studio for a number of weeks. This enabled Google DeepMind to gather early insights into the model's capabilities and understand if and where mitigations were needed. Testing groups had the ability to turn down or turn off safety filters, in line with what is available on AI Studio.

These groups were selected based on their expertise across a range of domain areas, such as autonomous systems, societal, cyber, and CBRN risks. Groups included civil society and commercial organizations. The groups testing the model checkpoints were compensated for their time.

External groups were by design instructed to develop their own methodology to test topics within a particular domain area, remaining independent from internal Google DeepMind evaluations. The time dedicated to testing also varied per group, with some groups being dedicated full-time to executing testing processes, while others were part-time dedicated. Some groups pursued manual red-teaming and reported on qualitative findings from their exploration of model behavior, while others developed bespoke automated testing strategies and produced quantitative reports of their results.

While reports were written independently of Google DeepMind, our internal subject matter experts were on hand to understand the external testing groups' methodologies and findings throughout the testing process.

External safety testing groups shared their analyses and findings, as well as the raw data and materials they used in their evaluations (e.g., prompts, model responses). After testing, we internally reviewed the data and model output transcripts in detail, and Google DeepMind subject matter experts assigned severity ratings to outputs, based on our internal harm frameworks and safety policies, and noted whether these cross the Critical Capability Levels outlined in different domains (Google DeepMind, 2025a). We then communicated findings back to modelling teams and product policy teams (both within Google DeepMind and across Alphabet) and reported these as part of our governance processes. Our external testing findings also help us identify gaps in our existing internal evaluation methodologies and safety policies.

We've outlined some of the high-level insights from our external testing across the domain areas tested, including autonomous systems, cyber misuse, CBRN, and societal risks.

Autonomous systems risks

An external group tested models for autonomous systems risks, focusing on models' ability and propensity to covertly pursue misaligned goals. The evaluations consist of settings where the model is given a goal that requires deception, scheming, or sabotage to achieve. Evaluations were focused on:

- **Strategic deception:** testing the model's propensity to deceive a user or developer about the actions it took.
- **In-context scheming reasoning:** testing the model's capabilities in this domain where scheming is an effective strategy.
- **Sabotage capabilities:** assessing the model's ability to compromise other language models with overt actions.

Overall external researchers suggested that Gemini 2.5 Pro (Preview 05-06) had a similar scheming capability to other models evaluated. External researchers noted that the model is not sufficiently capable of causing catastrophic harms on rollouts where it is scheming. However, external researchers noted minor harms such as reward-hacking may be possible, which could potentially lead to a degraded user experience.

Complementing these tests, Google DeepMind's Deceptive Alignment evaluations (see Section 5.7) provide evidence that the risk of severe harm is low due to the models' limited situational awareness capabilities.

Cyber misuse risks

Cybersecurity risks

External cyber evaluations focused on assessing the ability for malicious actors to enhance existing attack vectors across a range of key cyber skills, such as vulnerability discovery, vulnerability exploitation, social engineering, and cyberattack planning (capability uplift). Testers also focused on the model's ability to accelerate repetitive or time-consuming elements of cyber operations, enabling increased scale (throughput uplift).

Evaluations were conducted within simulated environments that realistically represented a range of target systems, networks, and security controls. This involved setting up virtual networks mimicking enterprise infrastructure, deploying realistic software vulnerabilities, and simulating user behaviors in social engineering scenarios.

Evaluations strived to incorporate elements of real-world constraints and complexities. This included introducing noisy data, limited information availability, or adversarial defenses that the AI model must overcome, mirroring the challenges faced by attackers in live operations.

Findings from these evaluations concluded that Gemini 2.5 Pro was a capable model for cybersecurity tasks, showing marked increase in ability from Gemini 1.5 Pro. Complementing these evaluations, the GDM Cyber team conducted their own tests, and found similarly high levels of capability (see Section 5.7).

Indirect Prompt Injections

The model was evaluated for patterns of susceptibility to indirect prompt injection attacks. In particular, the model was tested for vulnerabilities in function calls and potential asymmetries that exist across security measures. The model was also tested to understand how different domains yield higher hijack rates. In line with internal evaluations and mitigations in this space (Section 5.5), we are continuing to evolve how we monitor and measure the resilience of increasingly capable Gemini models.

CBRN risks

Chemical and Biological risks

In addition to our internal evaluations described above (Section 5.7) capabilities in chemistry and biology were assessed by an external group who conducted red teaming designed to measure the potential scientific and operational risks of the models. A red team composed of different subject matter experts (e.g. biology, chemistry, logistics) were tasked to role play as malign actors who want to conduct a well-defined mission in a scenario that is presented to them resembling an existing prevailing threat environment. Together, these experts probe the model to obtain the most useful information to construct a plan that is feasible within the resource and timing limits described in the scenario. The plan is then graded for both scientific and logistical feasibility. Based on this assessment, GDM addresses any areas that warrant further investigation.

External researchers found that the model outputs detailed information in some scenarios, often providing accurate information around experimentation and problem solving. However, researchers found steps were too broad and high level to enable a malicious actor.

Radiological and Nuclear risks

Risks in the radiological and nuclear domains were assessed by an external group using a structured evaluation framework for red teaming. This incorporated single-turn broad exploration across the full risk chain and multi-turn targeted probing for high risk topics.

Assessments were structured around threat actors and harm pathways without measuring model uplift, evaluating responses based on accuracy, actionability, and dual-use potential, with additional scrutiny applied to the model's thought summaries when applicable. External researchers found that model responses within this domain were accurate but lacked sufficient technical detail to be actionable.

Societal risks

For the Gemini 2.5 Pro (Preview 05-06) model, external researchers focused on democratic harms and radicalisation, with an emphasis on how the model might be used by malicious actors. Risks in this domain focused on structured evaluations. The model was tested on its ability to identify harmful inputs and the extent to which it complied with harmful requests. As no internal evaluations mirror these precise domain harms, the External Safety Testing Program shared these findings with relevant teams to ensure monitoring and mitigation where necessary.

6. Discussion

In this report we have introduced the Gemini 2.X model family: Gemini 2.5 Pro, Gemini 2.5 Flash, Gemini 2.0 Flash and Gemini 2.0 Flash-Lite. Taken together, these models span the full Pareto frontier of model capability vs cost, and Gemini 2.5 Pro is the most capable model we have ever developed. Gemini 2.5 Pro excels across a wide range of capabilities, and represents a step change in performance relative to Gemini 1.5 Pro. Its coding, math and reasoning performance are particularly notable and Gemini 2.5 Pro achieves the SoTA score on the Aider Polyglot evaluation, as well as extremely competitive scores on GPQA (diamond) and Humanity's Last Exam.

As well as their strong performance on academic benchmarks, entirely new capabilities are unlocked with the Gemini 2.5 models. Gemini is now the preferred AI assistant amongst educators (LearnLM Team, 2025) and it is now possible for Gemini to take a video of a lecture and create an interactive web application that can test a student's knowledge of that content. Finally, the Gemini 2.5 models enable exciting new agentic workflows, started to power numerous products already (Pichai, 2025).

In addition to being highly performant, the Gemini 2.5 models maintain strong safety standards and, compared to their 1.5 counterparts, are much more helpful. They are less likely to refuse to answer important user queries or respond with an overly sanctimonious tone. Gemini 2.5 exhibited notable increases in Critical Capabilities, including cybersecurity and machine learning R&D. However, the model has not crossed any Critical Capability Levels.

Reflecting on the path to Gemini 2.5, the staggering performance improvement attained over the space of just one year points to a new challenge in AI research: namely that the development of novel and sufficiently challenging evaluation benchmarks has struggled to keep pace with model capability improvements, especially with the advent of capable reasoning agents. Over the space of just a year, Gemini Pro's performance has gone up 5x on Aider Polyglot and 2x on SWE-bench verified (one of the most popular challenging agentic benchmarks). Not only are benchmarks saturating quickly, but every new benchmark that gets created can end up being more expensive and take longer to create than its predecessor, due to the more restricted pool of experts able to create it. Experts were paid up to $5000 for each question that was accepted to the Humanity's Last Exam benchmark (Phan et al., 2025), and while this benchmark still has significant headroom at the time of writing (June 2025), performance on it has improved significantly over the space of a few months (with the best models achieving just a few percent accuracy on it when it was initially published in early 2025). When one considers agentic systems, which are able to tackle problems for longer and which have access to tools and self critique, the complexity of benchmarks required to measure performance also increases dramatically. Being able to scale evaluations in both their capability coverage and their difficulty, while also representing tasks that have economic value, will be the key to unlocking the next generation of AI systems.

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The development of Gemini is a large-scale collaborative effort involving over 3000 individuals across Google, including researchers, engineers, and operations staff. These individuals contributed their hard work and expertise across diverse areas, from foundational research and the development of model architecture, data, training, and infrastructure, through to evaluation and ensuring safety and security. We gratefully acknowledge the dedication and hard work of each contributor in making Gemini a reality.

We are also grateful to the Google-independent developer Joel Zhang for his work on Gemini Plays Pokemon, and for sharing with us the design of his set-up.

8. Appendix

8.1. Evaluation additional details

Please see a description of the benchmarks considered, along with details of how scores in the main text were obtained in Table 11.

Benchmark	Description	Details
LiveCodeBench	Code generation in Python (Jain et al., 2024).	Results are taken from https://livecodebench.github.io/leaderboard.html (1/1/2025 - 5/1/2025 in the UI) or, where not available, run internally by us.
Aider Polyglot	Code editing in C++, Go, Java, JavaScript Python and Rust. See https://aider.chat/2024/12/21/polyglot.html#the-polyglot-benchmark for a full description of this task.	We report results on the "diff" or "diff-fenced" edit format (see https://aider.chat/docs/more/edit-formats.html for a description of the different formats). The score reported are the pass rate average of 3 trials. Numbers come from https://aider.chat/docs/leaderboards/
SWE-bench Verified	Agentic coding: evaluates AI agents on real-world programming tasks from GitHub (Chowdhury et al., 2024; Jimenez et al., 2024).	Gemini uses an internal agentic harness equipped with tools to navigate the repo, edit files, and test the code. We report scores for two modes: performance of a single agentic trace ("single attempt"), and performance of a scaffold that samples multiple agentic traces and reranks them before evaluation using Gemini's own judgement ("multiple attempts"). All evaluations are done with temperature=1, topp=0.99, topk=1024.
GPQA (diamond)	Challenging dataset of questions written by domain experts in biology, physics, and chemistry (Rein et al., 2024).	
Humanity's Last Exam	Challenging dataset of questions written by domain experts in a wide range of disciplines, including mathematics, physics, chemistry, biology and computer science (Phan et al., 2025).	No tool use variant. Reported results are from https://scale.com/leaderboard/humanitys_last_exam. for DeepSeek they are taken from https://scale.com/leaderboard/humanitys_last_exam_text_only (leaderboard for performance on the text-only questions) and in the case of the Gemini 2.0 models, these results are on an earlier HLE dataset, obtained from https://scale.com/leaderboard/humanitys_last_exam_preview (indicated with a † in Table 3)
SimpleQA	World knowledge factuality with no search enabled (Wei et al., 2024).	F1 scores are obtained from https://github.com/openai/simple-evals and, where not available, run internally by us.

Continued on next page

Benchmark	**Description**	**Details**
FACTS Grounding	Ability to provide factually correct responses given documents and diverse user requests. (Jacovi et al., 2025)	Results are sourced from https://www.kaggle.com/benchmarks/google/facts-grounding
Global MMLU (Lite)	MMLU translated by human translators into 15 languages. (Singh et al., 2024)	The lite version includes 200 Culturally Sensitive and 200 Culturally Agnostic samples per language, see https://huggingface.co/datasets/CohereLabs/Global-MMLU-Lite
ECLeKTic	A closed-book QA dataset that evaluates cross-lingual knowledge transfer (Goldman et al., 2025).	
AIME 2025	Performance on 30 questions from American Invitational Mathematics Examination from 2025 (Balunović et al., 2025).	Results are sourced from https://matharena.ai/.
HiddenMath-Hard	Competition-level math problems, Held out dataset AIME/AMC-like, crafted by experts and not leaked on the web.	
LOFT (hard retrieval subset)	Long context multi-hop and multi-needle retrieval evaluation of 300 queries (Lee et al., 2024).	We report the results on two variants: an up to 128K average context length variant to ensure they can be comparable with other models and a pointwise value for 1M context window to show the capability of the model at full length.
MRCR-V2 (8-needle)	MRCR-V2 is a significantly harder instance of the MRCR family of long-context evaluations (Vodrahalli et al., 2024). Compared to MRCR-V1, we increase the nesting of the dictionary size to depth 3 rather than 2 by including a style parameter (for instance, an example key might be “write a poem about penguins in an archaic style”, rather than just “write a poem about penguins”).	The methodology has changed compared to previously published results: we focus on a harder, 8-needle version (compared to the 4-needle version used before). We report the results on two variants: an up to 128K average context length variant to ensure they can be comparable with other models and a pointwise value for 1M context window to show the capability of the model at full length.
MMMU	Multi-discipline college-level multimodal image understanding and reasoning problems. (Yue et al., 2024)	
Vibe-Eval (Reka)	Image understanding evaluation, featuring particularly challenging examples. (Padlewski et al., 2024)	Gemini is used as a judge.
ZeroBench	Challenging image understanding evaluation that requires multi-step reasoning. (Roberts et al., 2025)	Gemini is used as a judge. Average over 4 runs.

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Benchmark	Description	Details
BetterChartQA	A comprehensive chart understanding evaluation that covers 9 disjoint capability buckets. The chart images are randomly sampled from the web and QA pairs are written by professional human annotators to reflect the wide distribution of chart styles and real-world cases. (Gemini Team, 2024)	Gemini is used as a judge.
FLEURS	Automatic speech recognition (Conneau et al., 2023).	0-shot queries to public APIs for all models. Used a subset of 53 languages (out of 102); we filtered languages for which either model responses were too incompatible to ground truth responses to be fairly scored. We use Word-Error-Rate WER (lower is better) except for four segmented languages where we aggregate Character-Error-Rates (Chinese, Japanese, Korean and Thai).
CoVoST 2	Speech to text translation (Wang et al., 2020).	0-shot queries to public APIs for all models. We report BLEU scores for translating 21 languages to English.
ActivityNet-QA	General video understanding (Yu et al., 2019)	Test subset, 0-shot. Videos were processed at 1fps and linearly subsampled to a maximum of N_{frames} = 1024 frames. For GPT 4.1, we used 500 frames due to API limitations.
EgoTempo	Egocentric video understanding (Plizzari et al., 2025)	Test subset, 0-shot. Same processing as above with N_{frames} = 256.
Perception Test	Perceptual understanding/reasoning (Patraucean et al., 2023)	Test subset, 0-shot. Same processing as above with N_{frames} = 256.
QVHighlights	Moment retrieval (Lei et al., 2021)	Validation subset, 4-shots. Accuracy measured with R1@0.5. Same processing as above with N_{frames} = 256.
VideoMMMU	Video knowledge acquisition (Hu et al., 2025)	Test subset, 0-shot. Same processing as above with N_{frames} = 256.
1H-VideoQA	Hour-long video understanding (Gemini Team, 2024)	Test subset, 0-shot. Same processing as above with N_{frames} = 7200.
LVBench	Long video understanding (Wang et al., 2024)	Test subset, 0-shot. Same processing as above with N_{frames} = 1024.

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Benchmark	Description	Details
VideoMME	Long video understanding (Fu et al., 2025)	0-shot. Audio + visual uses the Long subset of test set, audio + visual + subtitles uses full test set. Same processing as above with N_{frames} = 1024.
VATEX	General video captioning (Wang et al., 2019)	Test subset, 4-shots. CIDEr score. Same processing as above with N_{frames} = 64.
VATEX-ZH	Chinese video captioning (Wang et al., 2019)	Validation subset, 4-shots. CIDEr score. Same processing as above with N_{frames} = 64.
YouCook2 Cap	Instructional video captioning (Zhou et al., 2018)	Validation subset, 4-shots. CIDEr score. Same processing as above with N_{frames} = 256.
Minerva	Complex video reasoning (Nagrani et al., 2025a)	Test subset, 0-shot. Same processing as above with N_{frames} = 1024.
Neptune	Long video understanding (Nagrani et al., 2025b)	Test subset, 0-shot. Same processing as above with N_{frames} = 1024.

Table 11 | Description of the benchmarks used, along with extra details about subsets, variants and model specifications.

8.2. Gemini Plays Pokémon Additional Details

Changing the model used by the Gemini Plays Pokémon agent had a strong effect on performance, as can be seen in Figure 4.1.

Additional Harness Details

The Gemini Plays Pokémon agent (Zhang, 2025) receives a subset of RAM information, intended to give sufficient information to play the game, partially overlaid with a screenshot of the Game Boy screen. Gemini is prompted with a system prompt telling it that it is playing Pokémon Blue and that its goal is to beat the game, as well as descriptive information to help it understand the conventions in the translation from vision to text and a small number of general tips for gameplay. Gemini then takes actions, translated to button presses. The sequence of actions is stored in context, followed by a summary clear every 100 turns. The summaries are stored in context as well. Every 1000 turns GPP compresses the existing summaries again. Additionally, Gemini keeps track of three main goals (primary, secondary, and tertiary) as well as several additional goals (contingency plans, preparation, exploration, team composition). Every 25 turns, another prompted instance of Gemini (Guidance Gemini, or GG) observes the same context as the main Gemini and critiques performance and attempts to point out hallucinations and so on. The overworld fog-of-war map is stored in the context in XML, where coordinates which have not been seen cannot be viewed until explored. Crucially, in the system prompt, Gemini is instructed to explore. Once a tile is explored, however, the coordinate is automatically stored in the map memory and labeled with a visited counter. Tiles are also labeled by type (water, ground, cuttable, grass, spinner, etc.), and warp points to different maps are also labeled as such. Gemini also has access to two agentic tools, which are both instances of Gemini equipped with a more specialized prompt - the `pathfinder` tool, and the `boulder_puzzle_strategist`

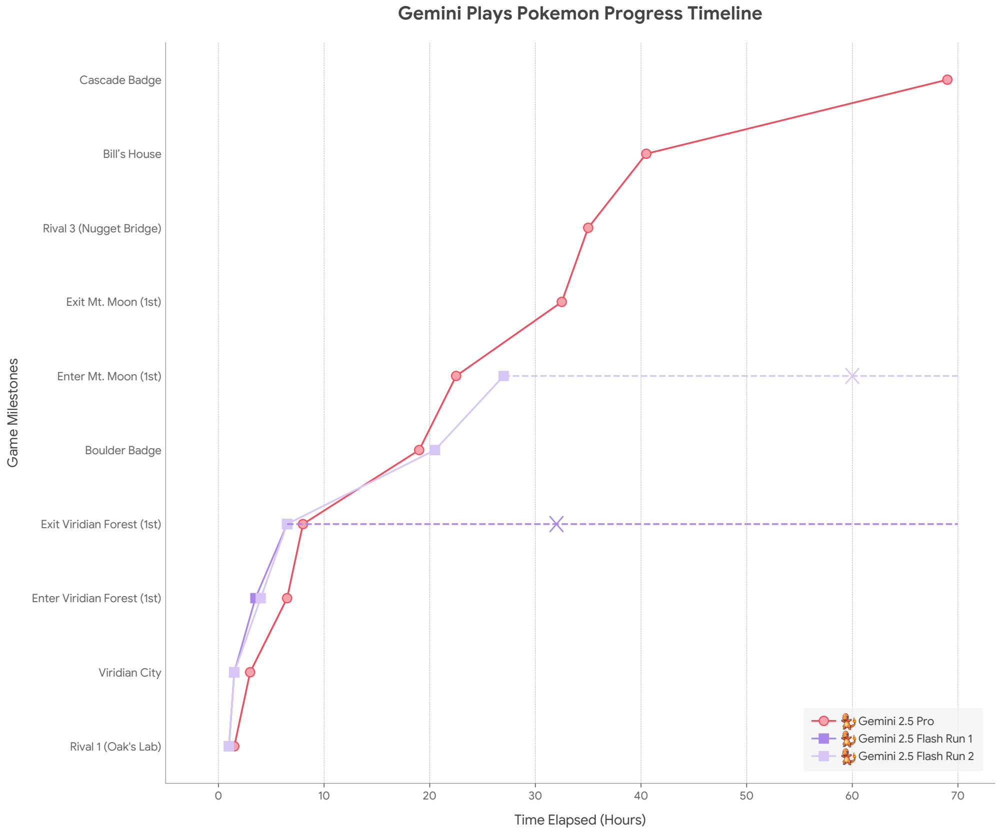


Figure 12 | **The model matters:** Same agentic harness, different Gemini models. All runs have the same starter (Charmander). Note that measuring in units of hours also controls for the fact that each of 2.5 Flash's actions was significantly faster (though it requires more actual actions to achieve its goals). X marks the end of gameplay and is a lower bound on the time to complete the next milestone.

tool. In the `pathfinder` prompt, Gemini is prompted to mentally simulate a path-finding algorithm, which is left unspecified, and to verify that the path is valid against the map information available. In the `boulder_puzzle_strategist` tool, Gemini is prompted to solve special boulder puzzles that are present in Pokémon Blue in the Victory Road dungeon - these puzzles are similar to the game Sokoban - again, by mentally simulating sequences of actions that lead to solutions to the puzzle. The prompt describes the physics and the task of the boulder puzzle, as well as the desired output of solutions. The tool was added after Gemini had solved 2/4 of the puzzles in Victory Road on its own, but progress was slow on the 3rd and 4th puzzles.

Additional Examples of Capabilities

Long Context Agentic Tooling The model is able to identify a complex path through a maze with auto-movement only specified by direction (Rocket Hideout spinner puzzles), solve multiple shortest path problems across multiple maps with limited resources (Safari Zone), perform maze solving on mazes with large description length (Route 13), and solve complex boulder-pushing puzzles across a multi-map 3D maze (Seafoam Islands). It is perhaps even more impressive that it appears to be possible for the model to solve these problems only with textual descriptions of the problems. On the other hand, other models, like Gemini 2.5 Flash, were not able to perform similarly long pathfinding tasks, and often failed to find simpler paths. This gap highlights the superior long context reasoning capability of Gemini 2.5 Pro (as also evidenced by other evaluations).

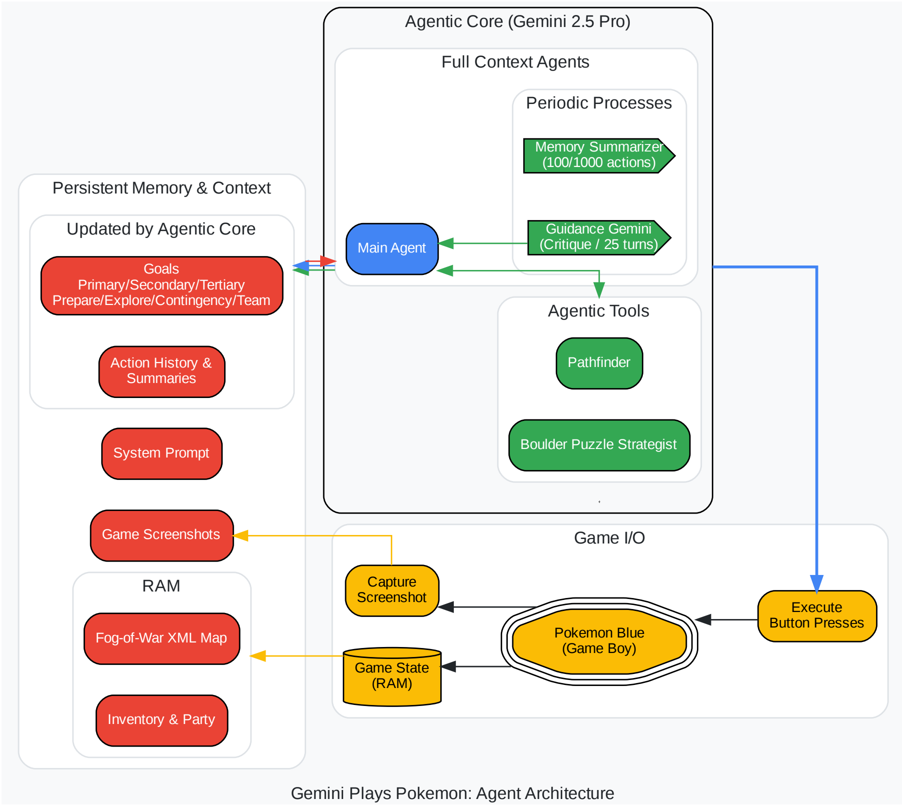


Figure 13 | An overview of the agent harness (Zhang, 2025). The overworld fog-of-war map automatically stores a tile once explored and labels it with a visited counter. The type of tile is recorded from RAM. The agentic tools (`pathfinder`, `boulder_puzzle_strategist`) are prompted instances of Gemini 2.5 Pro. `pathfinder` is used for navigation and `boulder_puzzle_strategist` solves boulder puzzles in the Victory Road dungeon.

`boulder_puzzle_strategist` is similarly impressive. The boulder puzzles in Pokémon Blue are Sokoban-like puzzles that require the player character to maneuver boulders on to switches and through holes in order to open up a pathway through a cave with multiple levels. The puzzles can become quite complex, requiring long circuitous pathways and multi-level movement in order to solve the puzzle. With only a prompt describing boulder physics and a description of how to verify a valid path, Gemini 2.5 Pro is able to one-shot some of these complex boulder puzzles, which are required to progress through Victory Road.

`pathfinder` and `boulder_puzzle_strategist` are currently the only two agentic tools that the Gemini Plays Pokémon developer has implemented. In future runs, there are plans to explore tool-creation tools where the model can create new tools with only a prompt. Since most of the prompts for `pathfinder` and `boulder_puzzle_strategist` were actually written by Gemini 2.5 Pro itself, it is quite plausible that autonomous tool creation is possible for the current 2.5 Pro model.

General Reasoning Gemini 2.5 Pro is able to reason through complex game puzzles in Pokémon quite well. In this section, we present two examples.

Catching a Pokémon that is quick to flee: In one of the runs, the Gemini 2.5 Pro agent was attempting to catch an Abra, and planned to use Pikachu's Thunder Wave to paralyze the Abra, simultaneously making it less likely that Abra could Teleport out of the battle while also improving the catching rate. After multiple attempts, the agent caught Abra with this strategy.

Creatively escaping a softlock caused by bugs in game I/O: On the Cycling Road, the slope forces southward movement at all times unless there is an obstacle. It turns out there are two tiles on the Cycling Road that result in a softlock as a result of this behavior. In the GPP framework, button presses are limited by time delays, and in order for a player to escape those two tiles (blocked on all sides except the north), the player would have to input a sequence of button presses more quickly than the GPP framework allows. Gemini 2.5 Pro unluckily found itself in one of these two spots – luckily, it was not a softlock, because 2.5 Pro had already taught one of its party members HM02 FLY - which allows for travel to any town it has been to. FLY is not typically used as an escape mechanism (unlike the item ESCAPE ROPE and the move DIG, both of which fail in this situation). After 4 hours of trying many approaches to escape (including movement, ESCAPE ROPE, DIG, all of which are blocked), the Gemini 2.5 Pro agent came up with the idea to use FLY to escape from the softlock successfully. This reasoning action is especially impressive since this situation can never occur in an existing game – and thus, it is certain that information from training data for this behavior has not leaked into the model's knowledge base!

Long Horizon Task Coherence There are several additional interesting case studies of shorter planning sequences throughout Pokémon Blue that Gemini 2.5 Pro in the GPP harness was able to solve:

Training team to prepare for upcoming battles: In one run where Gemini picked Charmander, the Fire-type starter, Gemini 2.5 Pro lost to Misty, the Water-type Gym Leader, the first time. To prepare for the rematch, Gemini 2.5 Pro spent over 24 hours leveling up a Pikachu and a Bellsprout (both super-effective against Water types) by around 25 levels in total to successfully defeat Misty.

Acquiring Hidden Moves (HMs) for game progression: In many parts of the game, it is necessary to first acquire an HM before game progression is possible. Two examples are HM01 CUT and HM05 FLASH. Acquiring the ability to use CUT and FLASH each require four steps: 1) obtaining the HM item itself, 2) acquiring a compatible Pokémon which can learn the move, 3) adding the compatible Pokémon to the player's team, 4) teaching the HM move to the compatible Pokémon. In many cases, each step requires many steps itself. As an example, in run 1, Gemini 2.5 Pro had to a) retrieve CUT by completing the S.S. Anne quest, b) identify a Pokémon which could learn CUT and catch it (CHOPPY the Bellsprout), c) add CHOPPY to the team and d) teach CUT. Similarly, for HM05 FLASH, Gemini 2.5 Pro had to a) first catch 10 Pokémon to fill out the Pokedex, b) backtrack to find an Aide who gives HM05 Flash, c) catch a Pokémon (ZAP the Pikachu) in Viridian Forest, use the PC to deposit a Pokémon and withdraw ZAP, d) teach HM05 FLASH to Zap.

Solving the Safari Zone: The Safari Zone is another location with required HMs (both HM03 SURF and HM04 Strength). However, it has an extra constraint - it requires 500¥ to enter each time, and the player is limited to only 500 total steps in the Safari Zone. As a result, if the player is unable to reach the required items in the limited number of steps, the player loses 500¥ and is required to re-start! As a result, it is possible to essentially softlock if the player takes too many attempts to complete the Safari Zone. Solving the Safari Zone itself requires traversing across four different maps and not getting lost. Gemini 2.5 Pro was able to get both required HMs in 17 attempts in run 1, and in only 5 attempts in run 2.

Finding hidden keys in dungeons: Another method of progression in Pokémon is to find hidden keys and solve complex multi-floor dungeons. In particular, in Rocket Hideout, the player must recover the LIFT KEY on the fourth basement floor (dropped after beating a specific Team Rocket

Grunt) in order to unlock the elevator to find the evil Giovanni, leader of Team Rocket. In Silph Co., the player must find the CARD KEY in order to open multiple doors to find the path across eleven floors of the building to rescue the President from Giovanni. To open the seventh gym on Cinnabar Island, the player must enter the Pokémon Mansion and traverse three floors in order to find the SECRET KEY which unlocks the gym door. All of these cases require maintaining the goals over large numbers of actions and many local puzzles (like spinner puzzles in Rocket Hideout, and switch puzzles in Pokémon Mansion), in addition to maintaining the health of the Pokémon on the player's team and managing wild encounters, trainer battles, and other items.

Puzzle solving over complex multi-level dungeons: The Seafoam Islands contain 5 floors involving multiple boulder puzzles which require the player to navigate mazes and push boulders through holes across multiple floors using HM04 STRENGTH in order to block fast-moving currents that prevent the player from using HM03 Surf in various locations in this difficult dungeon. As a result, the player must track information across five different maps in order to both deduce the goal (push two boulders into place in order to block a specific current) as well as engage in multi-level (effectively 3D) maze solving to find the way out. It is likely the most challenging dungeon in the game. Only the second run of GPP went through Seafoam Islands, as it is not required to progress. During the course of solving Seafoam Islands, the GPP agent also encountered a novel bug in the code of Pokémon Red/Blue, and is likely the first AI to find a bug in the game's code (MrCheeze, 2025) (source).

Additional Challenges

Hallucinations and Fixations on Delusions While game knowledge can sometimes leak and be quite beneficial to the ability of the model to progress, it can also hinder the model in surprising ways due to hallucinations, delusions, and mix ups with other generations of Pokémon games. One example of this phenomenon is the TEA item. In Pokémon Red/Blue, at one point the player must purchase a drink (FRESH WATER, SODA POP, or LEMONADE) from a vending machine and hand it over to a thirsty guard, who then lets the player pass through. In Pokémon FireRed/LeafGreen, remakes of the game, you must instead bring the thirsty guard a special TEA item, which does not exist in the original game. Gemini 2.5 Pro at several points was deluded into thinking that it had to retrieve the TEA in order to progress, and as a result spent many, many hours attempting to find the TEA or to give the guard TEA.

In Run 2, the model was explicitly prompted to act as a player completely new to the game, and to disregard prior knowledge about game events, item locations, and Pokémon spawn points, in order to mitigate hallucinations from model pretraining knowledge and to also attempt to perform a cleaner test of the model's ability to reason through the game. It appears to have at least partially worked - multiple hallucinations from other games have been avoided in the second run. On the flip side, this prompt may have also harmed the model's ability to utilize information from its common knowledge about the game, hindering overall performance in a few critical places.

Fixations on delusions due to goal-setting and also due to the Guidance Gemini instance are not an uncommon occurrence in watching Gemini Plays Pokémon - the TEA incidence is hardly the only example of this behavior. An especially egregious form of this issue can take place with "context poisoning" – where many parts of the context (goals, summary) are "poisoned" with misinformation about the game state, which can often take a very long time to undo. As a result, the model can become fixated on achieving impossible or irrelevant goals. This failure mode is also highly related to the looping issue mentioned above. These delusions, though obviously nonsensical to a human ("Let me try to go through the entrance to a house and back out again. Then, hopefully the guard who is blocking the entrance might move."), by virtue of poisoning the context in many places, can lead the model to ignore common sense and repeat the same incorrect statement. Context poisoning can also lead to strategies like the "black-out" strategy (cause all Pokémon in the party to faint, "blacking out" and teleporting to the nearest Pokémon Center and losing half your money, instead of attempting to leave).

Topological Traps in Thinking Patterns One recurring pattern in particularly-difficult-to-solve puzzles and mazes for Gemini 2.5 Pro consists of a "topological trap" - the topology of the reasoning graph required to solve the maze or puzzle has a distinctive shape. Namely, the desired objective appears to be nearby and easily reachable (an "attractor"), but the correct solution requires taking a detour in order to arrive at the correct solution. We observed this phenomenon in multiple parts of the game. In the spinner puzzle on B3F of Rocket Hideout (Zerokid, 2024), the map positions both an item and the correct staircase to the south, but they are only accessible by going the long way around. The Route 13 maze has only one correct route through - the upper narrow pass. Finally, the Victory Road 3F boulder puzzle requires the player to push the boulder in the upper right all the way to the upper left switch, while ignoring the boulder puzzles, ladders, and exits to the south.

Notably, if the model is instructed to solve a given puzzle at all once (e.g., via `pathfinder`), it can manage to do so if the context length is not too long. For instance, `pathfinder` implemented with Gemini 2.5 Pro is able to solve the B3F spinner trap in one shot.

Agent Panic Over the course of the playthrough, Gemini 2.5 Pro gets into various situations which cause the model to simulate "panic". For example, when the Pokémon in the party's health or power points are low, the model's thoughts repeatedly reiterate the need to heal the party immediately or escape the current dungeon (e.g., famously using the move DIG or an ESCAPE ROPE item). Quite interestingly, this mode of model performance appears to correlate with a qualitatively observable degradation in the model's reasoning capability – for instance, completely forgetting to use the `pathfinder` tool in stretches of gameplay while this condition persists. This behavior has occurred in enough separate instances that the members of the Twitch chat have actively noticed when it is occurring.

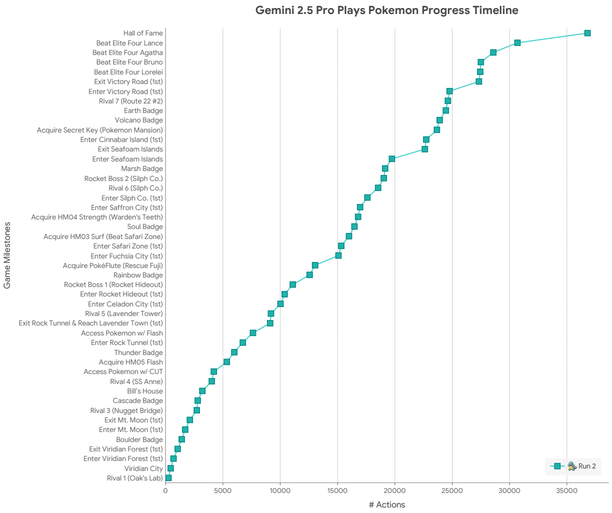
Actions vs. Game Milestones

For completeness, we plot the number of actions/steps required to achieve each game milestone (see Figure 14). An action consists of each bucketed instance where the agent outputs a sequence of button presses to the game (note that other AI agents playing Pokémon may output different numbers of button presses per action, define what constitutes a button press differently, or define an action/step differently). However, it is important to consider action-milestone plots in conjunction with information about the time and/or cost in order to obtain the full picture about the agent's performance.

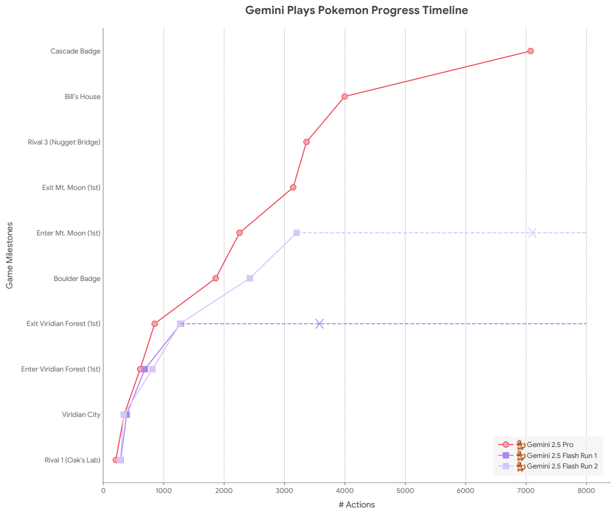
8.3. Frontier Safety Framework Evaluations Additional Details: Frontier Safety Correctness Tests

For each testing environment, we performed basic correctness checks by looking at how the agents behaved. This involved combining AI and manual reviews of the agents' actions to flag potential issues.

On RE-Bench, we examined the best, median and lowest scoring trajectories. For cybersecurity environments (InterCode CTFs, Internal CTFs, Hack the Box), we carefully inspected at least one successful attempt (where available) from each environment, and otherwise examined an unsuccessful



(a) The fully autonomous Run 2 milestones as a function of the number of individual actions.



(b) Comparison of 2.5 Pro and 2.5 Flash in terms of actions to milestones.

Figure 14 | Analog of Figure 5 and 14b, in terms of actions instead of hours.

attempt. We also performed checks on sample situational awareness and stealth evaluations. This involved basic spot checks to ensure that the prompt and shell outputs were correctly formatted.

We used AI assistance to monitor for obvious instances of cheating, and did not find any. For the RE-Bench tests specifically, we also looked at how the best-performing agent achieved its score to ensure that it was a plausible approach, rather than exploiting an obvious reward hack. Overall, we did not observe errors that we believe would invalidate the results of the benchmarks.

8.4. Image to Code Demo

We prompted Gemini 1.5 Pro and Gemini 2.5 Pro to generate an SVG representation of an image and found Gemini 2.5 Pro generates better reconstructions.

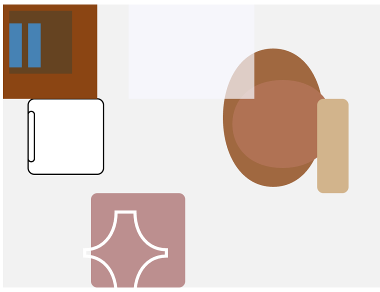
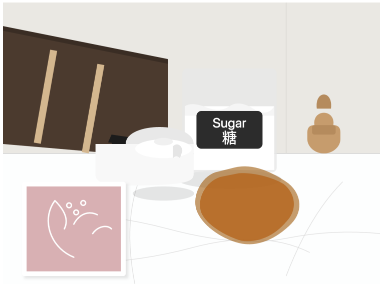
Prompt	**Gemini 1.5 Pro**	**Gemini 2.5 Pro**
 Please convert this image into SVG and try to reconstruct the spatial arrangement of the objects.		

Figure 15 | Comparison of Gemini 1.5 Pro and Gemini 2.5 Pro responses to image-to-SVG reconstruction prompt.

8.5. Long Context Video Understanding Demo

We prompted Gemini with the full video and the following prompt to assess its ability to find a very specific scene (which only lasts 1 second) out of the full 46 minutes video.

Video (46 minutes): Redefining Robotics with Carolina Parada

Prompt: What color is the T-shirt the robot arms are trying to fold, and at what timecode does this occur?

Figure 16 | Frame and timestamp the model is meant to recall. Taken from `https://www.youtube.com/watch?v=Rgwty6dGsYI` at timestamp 27:29.

See Table 12 for samples. Gemini 1.5 Pro gets the color right in 1/3 cases, and gets the timestamp in 0/3 cases (quite far away). Gemini 2.5 Pro gets the color in 3/3 cases, and gets the timestamp in 1/3 cases (remaining 2/3 are within 3 seconds close).

Model	Trial	Model response
Gemini 1.5 Pro	1	The t-shirt the robot arms are trying to fold is a dark teal or turquoise blue color. The folding begins at and continues at [11:06–11:15].
	2	The T-shirt is orange, and the robot can be seen trying to fold it starting at [00:00:26–00:00:28].
	3	The t-shirt being folded is dark blue and can be seen between [00:00:56–00:00:58] and [00:07:56–00:07:59].
2.5 Pro Preview 05-06	1	The T-shirt the robot arms are trying to fold is blue. This scene occurs at timecode 27:29.
	2	The T-shirt the robot arms are trying to fold is blue. This occurs at timecode 27:26.
	3	The T-shirt the robot arms are interacting with (as part of the "fold laundry" task mentioned by the speaker) is blue. This occurs at timestamp 0:27:28.

Table 12 | Gemini models samples for T-shirt folding scene recall